

NMFS EBS Shelf Survey Data

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0.1 NMFS EBS bottom trawl shelf survey

Time series of annual estimates of area-swept abundance and biomass, as well as size composition data, are available for PIBKC from the summer NMFS EBS Shelf Bottom Trawl Survey based on the stock area first defined in the 2013 assessment (Foy 2013), which includes the Pribilof District and a 20 nm strip adjacent to the eastern edge of the District (Figure ?@fig-RegAreaQ). The adjacent area was added as a result of the 2015 rebuilding plan and the concern that crab outside the Pribilof District were not being accounted for in the assessment. The survey has been conducted annually since 1975, with the exception of 2020. In 2020, the survey was not conducted due to issues associated with the global COVID-19 pandemic.

The standardized EBS bottom trawl survey is based on a systematic design with a fixed sampling station at the center of each 37.04×37.04 km (20×20 nautical mile) grid square (Lauth and Nichol 2013). In the area surrounding the Pribilof Islands, high-density “corner stations” are sampled to better assess local blue king crab concentrations (Figure 1). Since 1982, the survey has used standard 83-112 Eastern otter trawls, which have 25.3-m (83 ft) headropes and 34.1-m (112 ft) footropes, to sample crab and groundfish species at 77 stations within the Pribilof District, augmented by a column of 9 stations to the east of the District (indicated by the dashed red line in Figure ?@fig-RegAreaQ) to better encompass the stock limits. The standard tow is nominally 30 minutes on bottom at a tow speed of 3 knots (~ 1.5 nmi distance), but net mensuration gear is used to more accurately assess time and distance “on bottom” as well as net width to provide a precise estimate of area swept. The net mensuration gear also allows the collection of depth and temperature data. Details of the NMFS bottom trawl protocols established by the National Oceanic and Atmospheric Administration can be found in (Stauffer 2004).

For each tow, all crab were removed from the catch, sorted by species and sex, and a total catch weight was obtained for each species (e.g., Zacher et al. 2023). All blue king crab were sampled for biological characteristics, including sex, carapace length (to 0.1 mm), weight, shell condition, and egg color, egg condition, and clutch size for females. Male crab were characterized as immature, mature, sublegal, and legal based on the size categories in Table 10. Females were characterized as immature or mature based on abdominal flap morphology and egg presence (Zacher et al. 2023).

Biomass estimates were calculated using the number of individual male and female crab at each 1 mm size category, using weight-size relationships developed by the AFSC’s Kodiak Laboratory (the same as those applied to fishery data: Equation ?@eq-WatZ; Zacher et al. (2023)). Weights were calculated for each 1 mm size bin and summed within the legal male, sublegal male, mature, and immature size categories for each sex caught at a station. Total biomass was estimated by

averaging crab density (biomass /area swept) from all stations within the area-augmented Pribilof District, and multiplying by the total area (Zacher et al. 2023).

Forty-five stations were included in survey strata for PIBKC in 1975, increasing to 86 by 1983 and remaining essentially constant since then (Tables 11 and 12). In the early 1980s, males were found at up to 38 of these stations and females were found at up to 24. This decreased in the 1990s when males occurred in a maximum of 22 stations, with females occurring at a maximum of 15 stations. Since 2010, the maximum number of stations at which males were caught is 9, with a median of 5, while females were caught at a maximum of 8 stations, with a median of 4. In similar fashion, the number of males caught declined from a maximum of 858 in 1975 to a since-2010 maximum of 22; for females, the corresponding numbers are 343 (in 1981) and 24. In most years, more mature crab were caught than immature, although there were exceptions (e.g., 1989 for both sexes). In 2023, a total of 2 (immature/sublegal) males and 7 (mature) females were caught at 2 and 1 stations, respectively, all in the high-density sampling area (Tables 11 and 12). No mature males were caught in 2023. In 2024, no PIBKC were caught.

Annual survey abundance and biomass for PIBKC have declined precipitously over the course of the 45 year time series (Tables 13-18 and Figures 2-13). On decadal scales, mean survey abundance and biomass have declined for males from 13.141 million crab and 29.53 thousand t in the 1970s to 0.224 million crab and 0.402 thousand t in the 2010s. Similarly, mean survey abundance and biomass have declined for females from 8.862 million crab and 8.078 thousand t in the 1970s to 0.255 million crab and 0.230 thousand t in the 2010s. Dampened oscillations in survey abundance and biomass have occurred on roughly decadal scales for this stock, with maxima exhibited at the start of the time series for males, followed by a decline to low values in the mid-to-late 1980s, an increase to a relative maximum in the early 1990s, followed by a decline to consistent low values since 1999 (a “blip” with large confidence intervals in 2005 was the exception). Females show a similar pattern, but lagged perhaps 5 years or so (without a “blip” in 2005). In 2019, apparent increases observed in mature and legal male biomass estimates relative to 2018 were attributed primarily to an abbreviated, but “still valid,” tow that may have had the effect of artificially increasing the CPUE calculated for the affected station (Zacher et al. 2020).

One feature that characterizes survey-based estimates of abundance and biomass for PIBKC is the large uncertainty (cv on the order of 0.5-1) associated with the estimates, which complicates the interpretation of sometimes large interannual swings in estimates of abundance (Tables 15 and 16, Figures 2-7) and biomass (Tables 17 and 18, Figures 8-13). Estimated total abundance of male PIBKC from the NMFS EBS bottom trawl survey declined from ~24 million crab in 1975, the first year of the “standardized” survey, to ~150,000 in 2016 (the lowest estimated abundance since 2004, which was the minimum for the time series. Following a general decline to a low-point in 1985 (~500,000 males), abundance increased by a factor of 10 in the early 1990s, then generally declined (with small-amplitude oscillations superimposed) to the present. Estimated female abundance generally followed a similar trend, spiking at 180 million crab in 1980, from ~13 million crab in 1975 and only ~1 million in 1979, then returned to more typical levels in 1981 (~6 million crab). More recently, abundance has fluctuated around 200,000 females. Estimated biomass for both males and females has followed trends similar to those in abundance.

Size frequencies across the entire time series are shown by sex in Figures 14-16. Based on patterns for crab > 50 mm CL, a single recruitment event starting in 1988 is evident in Figure 15, with a second possible event starting in 2005. However, these plots provide little evidence of recent recruitment.

Table 1

Table 2

The small numbers of crab caught in recent surveys make it difficult to draw firm conclusions regarding spatial patterns (Figures 17-20). Examining decadal-averaged patterns, however, there appears to have been a fairly strong contraction in range from extending beyond the PIHCZ in the 1980s to contained within the PIHCZ currently. The current spatial pattern of PIBKC abundance is centered fairly compactly within the Pribilof District to the east of St. Paul Island and north of St. George Island, within a 60 nm radius of St. Paul.

Table 3

Table 4

Table 5

Table 6

Table 7

Table 8

Table 9

Tables

List of Tables

1		3
2		3
3		3
4		4
5		4
6		4
7		4
8		4
9		4
10	Size groups for various male components of the PIBKC stock used here. Female maturity is based on abdominal flap morphology and egg presence.	5
11	Sample sizes (number of survey hauls, number hauls where crab were caught, number of crab caught) for male population components in the NMFS EBS trawl survey in the Pribilof District.	6
12	Sample sizes (number of survey hauls, number hauls where crab were caught, number of crab caught) for female population components in the NMFS EBS trawl survey in the Pribilof District.	8
13	Summary statistics for trawl survey abundance by decade, in millions.	10
14	Summary statistics for trawl survey biomass by decade, in 1,000s t.	11
15	Estimated annual abundance (millions of crab) of male PIBKC population components from the NMFS EBS trawl survey.	12
16	Estimated annual abundance (millions of crab) of female PIBKC population components from the NMFS EBS trawl survey.	14
17	Estimated annual biomass (1,000s t) of male PIBKC population components from the NMFS EBS trawl survey.	16
18	Estimated annual biomass (1,000s t) of female PIBKC population components from the NMFS EBS trawl survey.	18

Table 10. Size groups for various male components of the PIBKC stock used here. Female maturity is based on abdominal flap morphology and egg presence.

sex	size.range	category
male	< 120 mm CL	immature male
male	> 119 mm CL	mature male
male	< 135 mm CL	sublegal male
male	> 134 mm CL	legal male

Table 11. Sample sizes (number of survey hauls, number hauls where crab were caught, number of crab caught) for male population components in the NMFS EBS trawl survey in the Pribilof District.

year	survey	immature males		mature males		sublegal males		legal males		all males	
	number of hauls	non-0 hauls	no. crab	non-0 hauls	no. crab	non-0 hauls	no. crab	non-0 hauls	no. crab	non-0 hauls	no. crab
1975	45	11	305	13	553	11	530	13	328	13	858
1976	59	3	105	11	91	9	122	10	74	12	196
1977	58	7	56	10	129	9	73	9	112	10	185
1978	58	8	60	11	130	10	112	10	78	12	190
1979	33	2	2	9	77	6	23	9	56	9	79
1980	70	10	41	21	133	12	64	21	110	21	174
1981	84	19	99	36	184	23	128	36	155	38	283
1982	84	19	70	35	114	21	84	31	100	38	184
1983	86	15	47	32	93	18	74	29	66	35	140
1984	86	10	27	20	37	17	37	16	27	25	64
1985	86	3	4	14	24	8	13	11	15	14	28
1986	86	1	1	13	26	2	2	13	25	13	27
1987	86	5	34	15	50	6	38	14	46	16	84
1988	85	5	52	5	12	5	52	5	12	9	64
1989	86	8	160	4	11	8	160	4	11	10	171
1990	86	8	90	10	59	11	126	7	23	14	149
1991	85	16	92	19	103	20	129	14	66	22	195
1992	86	12	89	14	73	13	119	12	43	17	162
1993	85	12	75	19	96	15	115	17	56	21	171
1994	86	8	32	18	68	12	51	18	49	19	100
1995	86	7	66	18	177	15	118	14	125	19	243
1996	86	7	32	19	87	11	54	19	65	20	119
1997	86	7	25	17	65	10	39	16	51	19	90
1998	85	12	56	20	56	15	66	17	46	21	112
1999	86	7	9	13	34	9	18	11	25	15	43
2000	85	4	9	16	40	9	20	13	29	16	49
2001	86	3	5	6	28	4	9	5	24	7	33
2002	86	0	0	6	12	1	1	6	11	6	12
2003	86	2	2	7	14	3	3	7	13	9	16
2004	85	3	5	3	3	5	7	1	1	6	8
2005	84	3	54	2	5	3	54	2	5	4	59
2006	86	4	7	3	3	4	8	2	2	6	10
2007	86	4	14	2	6	4	17	2	3	4	20
2008	86	2	13	1	1	2	13	1	1	3	14
2009	86	5	16	3	15	5	27	3	4	5	31
2010	86	2	6	5	8	3	10	4	4	5	14
2011	86	0	0	3	9	2	2	2	7	3	9
2012	86	1	9	4	13	1	14	4	8	4	22
2013	86	1	3	2	6	2	5	2	4	3	9
2014	86	3	5	2	5	3	5	2	5	4	10
2015	86	2	4	8	13	6	10	5	7	9	17

(continued)

year	survey	immature males		mature males		sublegal males		legal males		all males	
	number of hauls	non-0 hauls	no. crab	non-0 hauls	no. crab	non-0 hauls	no. crab	non-0 hauls	no. crab	non-0 hauls	no. crab
2016	86	4	5	3	3	5	7	1	1	5	8
2017	86	2	4	4	4	3	5	3	3	5	8
2018	86	4	6	3	3	4	6	3	3	5	9
2019	86	5	8	3	3	5	8	3	3	6	11
2021	86	1	1	5	9	3	4	4	6	5	10
2022	86	0	0	2	2	0	0	2	2	2	2
2023	86	2	2	0	0	2	2	0	0	2	2
2024	70	0	0	0	0	0	0	0	0	0	0

Table 12. Sample sizes (number of survey hauls, number hauls where crab were caught, number of crab caught) for female population components in the NMFS EBS trawl survey in the Pribilof District.

year	survey	immature females		mature females		all females	
	number of hauls	non-0 hauls	no. crab	non-0 hauls	no. crab	non-0 hauls	no. crab
1975	45	0	0	9	265	9	265
1976	59	3	81	4	11	5	92
1977	58	2	9	5	136	5	145
1978	58	1	1	8	107	8	108
1979	33	2	3	4	22	5	25
1980	70	3	6	11	337	11	343
1981	84	13	31	20	202	23	233
1982	84	5	35	23	264	24	299
1983	86	6	15	17	288	18	303
1984	86	6	24	14	145	15	169
1985	86	7	15	8	28	12	43
1986	86	2	2	8	106	10	108
1987	86	5	22	7	36	11	58
1988	85	5	38	8	20	9	58
1989	86	8	131	9	40	13	171
1990	86	5	75	9	90	10	165
1991	85	9	36	11	126	15	162
1992	86	4	66	9	76	11	142
1993	85	5	45	13	89	15	134
1994	86	3	8	12	271	13	279
1995	86	3	38	11	220	12	258
1996	86	7	13	10	213	12	226
1997	86	4	17	11	137	13	154
1998	85	8	29	11	107	15	136
1999	86	0	0	10	155	10	155
2000	85	0	0	13	74	13	74
2001	86	1	1	9	93	10	94
2002	86	1	1	6	66	7	67
2003	86	4	4	7	69	9	73
2004	85	3	5	3	4	5	9
2005	84	1	43	5	15	6	58
2006	86	4	6	3	22	6	28
2007	86	3	7	2	9	5	16
2008	86	3	19	4	24	6	43
2009	86	3	9	3	29	4	38
2010	86	5	9	4	15	7	24
2011	86	1	1	2	2	3	3
2012	86	1	1	5	15	6	16
2013	86	2	2	4	8	5	10
2014	86	1	1	3	4	4	5
2015	86	0	0	4	11	4	11

(continued)

year	survey	immature females		mature females		all females	
	number of hauls	non-0 hauls	no. crab	non-0 hauls	no. crab	non-0 hauls	no. crab
2016	86	4	5	7	19	8	24
2017	86	4	5	4	10	6	15
2018	86	1	1	3	6	4	7
2019	86	0	0	2	11	2	11
2021	86	0	0	3	12	3	12
2022	86	0	0	4	7	4	7
2023	86	0	0	1	7	1	7
2024	70	0	0	0	0	0	0

Table 13. Summary statistics for trawl survey abundance by decade, in millions.

category	1970		1980		1990		2000		2010		decade 2020	
	mean	max	mean	max	mean	max	mean	max	mean	max	mean	max
immature females	1.706	7.369	0.7645	2.636	0.756	2.177	0.3201	2.2681	0.05116	0.1656	0.00000	0.00000
mature females	7.156	13.880	21.3116	182.903	3.008	5.047	0.7272	1.6975	0.20400	0.3594	0.11685	0.22932
all females	8.862	14.732	22.0762	183.684	3.764	5.322	1.0472	2.5573	0.25516	0.4544	0.11685	0.22932
immature males	4.042	8.476	1.3213	3.515	1.237	2.450	0.3257	1.9813	0.09662	0.1945	0.01317	0.03322
mature males	9.099	15.288	1.8942	7.842	1.619	3.102	0.2274	0.7251	0.12712	0.2722	0.05210	0.17362
sublegal males	6.497	14.712	1.6675	4.331	1.791	3.349	0.3850	1.9813	0.13763	0.3026	0.02788	0.07831
legal males	6.644	11.769	1.5480	6.244	1.065	2.186	0.1681	0.5276	0.08610	0.1642	0.03738	0.11475
all males	13.141	23.764	3.2155	10.575	2.856	4.371	0.5531	2.0733	0.22373	0.4668	0.06527	0.19306

Table 14. Summary statistics for trawl survey biomass by decade, in 1,000s t.

category	1970		1980		1990		2000		2010		decade 2020	
	mean	max	mean	max	mean	max	mean	max	mean	max	mean	max
immature females	1.125	4.968	0.3149	0.8008	0.3763	1.118	0.09232	0.4773	0.02422	0.08408	0.000000	0.00000
mature females	6.953	13.154	24.4680	211.6037	2.9518	5.408	0.81884	1.8163	0.20584	0.41163	0.131925	0.26241
all females	8.078	13.572	24.7829	212.3032	3.3281	5.585	0.91115	1.8167	0.23006	0.41163	0.131925	0.26241
immature males	3.811	8.341	0.7711	2.0838	0.9836	2.004	0.13309	0.3258	0.07633	0.16471	0.009729	0.02392
mature males	25.721	42.618	5.7347	23.5529	4.0885	8.360	0.65383	2.0913	0.32571	0.64394	0.129180	0.40462
sublegal males	8.148	19.378	1.3954	4.9581	1.9477	3.567	0.23745	0.5649	0.14687	0.34967	0.036506	0.12211
legal males	21.383	40.366	5.1104	20.6786	3.1245	6.787	0.54947	1.7457	0.25518	0.45898	0.102403	0.29751
all males	29.532	46.395	6.5058	25.6367	5.0721	9.328	0.78692	2.2047	0.40204	0.80865	0.138909	0.41962

Table 15. Estimated annual abundance (millions of crab) of male PIBKC population components from the NMFS EBS trawl survey.

	immature males		mature males		sublegal males		legal males		all males	
year	est.	cv	est.	cv	est.	cv	est.	cv	est.	cv
1975	8.476	0.567	15.288	0.502	14.712	0.479	9.051	0.501	23.764	0.466
1976	4.960	0.954	4.782	0.445	5.729	0.882	4.012	0.471	9.742	0.589
1977	4.216	0.457	13.044	0.743	5.491	0.440	11.769	0.771	17.260	0.625
1978	2.421	0.502	6.141	0.496	4.639	0.419	3.923	0.616	8.562	0.428
1979	0.139	0.699	6.240	0.360	1.913	0.472	4.467	0.347	6.380	0.357
1980	2.733	0.466	7.842	0.408	4.331	0.458	6.244	0.420	10.575	0.400
1981	2.099	0.324	3.834	0.180	2.688	0.317	3.246	0.177	5.934	0.207
1982	1.371	0.281	2.354	0.181	1.654	0.255	2.071	0.188	3.725	0.172
1983	1.031	0.357	1.851	0.186	1.561	0.309	1.321	0.170	2.882	0.220
1984	0.518	0.397	0.771	0.225	0.730	0.290	0.558	0.247	1.288	0.212
1985	0.068	0.598	0.428	0.281	0.226	0.340	0.270	0.294	0.496	0.269
1986	0.019	1.000	0.480	0.305	0.039	0.698	0.460	0.313	0.499	0.298
1987	0.622	0.834	0.903	0.414	0.695	0.748	0.830	0.416	1.525	0.434
1988	1.238	0.842	0.238	0.509	1.238	0.842	0.238	0.509	1.476	0.708
1989	3.515	0.588	0.240	0.624	3.515	0.588	0.240	0.624	3.755	0.585
1990	2.450	0.596	1.470	0.626	3.349	0.596	0.572	0.538	3.920	0.578
1991	1.920	0.373	2.014	0.363	2.697	0.332	1.238	0.444	3.935	0.343
1992	2.436	0.588	1.935	0.420	3.217	0.520	1.154	0.453	4.371	0.475
1993	1.484	0.520	1.876	0.310	2.245	0.432	1.114	0.300	3.359	0.339
1994	0.639	0.374	1.294	0.341	0.998	0.343	0.935	0.345	1.933	0.332
1995	1.147	0.889	3.102	0.600	2.062	0.744	2.186	0.615	4.249	0.675
1996	0.719	0.625	1.712	0.281	1.162	0.547	1.269	0.263	2.431	0.334
1997	0.467	0.525	1.201	0.294	0.736	0.464	0.933	0.284	1.669	0.342
1998	0.949	0.458	0.967	0.246	1.119	0.414	0.797	0.253	1.917	0.309
1999	0.160	0.373	0.617	0.334	0.324	0.388	0.453	0.345	0.777	0.327
2000	0.164	0.563	0.725	0.296	0.361	0.385	0.528	0.297	0.889	0.312
2001	0.093	0.645	0.522	0.710	0.169	0.595	0.446	0.744	0.615	0.690
2002	0.000	0.000	0.225	0.473	0.018	1.000	0.207	0.495	0.225	0.473
2003	0.045	0.717	0.229	0.389	0.061	0.589	0.214	0.402	0.274	0.341
2004	0.088	0.590	0.048	0.563	0.120	0.460	0.016	1.000	0.136	0.417

(continued)

	immature males		mature males		sublegal males		legal males		all males	
year	est.	cv	est.	cv	est.	cv	est.	cv	est.	cv
2005	1.981	0.964	0.092	0.712	1.981	0.964	0.092	0.712	2.073	0.921
2006	0.138	0.495	0.056	0.564	0.155	0.503	0.038	0.699	0.194	0.419
2007	0.246	0.717	0.110	0.854	0.302	0.644	0.054	0.745	0.356	0.639
2008	0.234	0.928	0.018	1.000	0.234	0.928	0.018	1.000	0.252	0.862
2009	0.268	0.631	0.249	0.732	0.448	0.697	0.068	0.588	0.516	0.676
2010	0.101	0.841	0.130	0.486	0.167	0.728	0.065	0.482	0.232	0.608
2011	0.000	0.000	0.166	0.792	0.036	0.698	0.129	0.868	0.166	0.792
2012	0.195	1.000	0.272	0.797	0.303	1.000	0.164	0.678	0.467	0.879
2013	0.076	1.000	0.104	0.862	0.112	0.745	0.069	0.804	0.181	0.644
2014	0.091	0.591	0.092	0.710	0.091	0.591	0.092	0.710	0.183	0.566
2015	0.076	0.766	0.234	0.367	0.185	0.525	0.125	0.446	0.309	0.408
2016	0.094	0.517	0.056	0.563	0.131	0.458	0.019	1.000	0.150	0.488
2017	0.068	0.773	0.091	0.503	0.087	0.637	0.072	0.589	0.159	0.456
2018	0.110	0.572	0.056	0.563	0.110	0.572	0.056	0.563	0.166	0.521
2019	0.155	0.485	0.071	0.575	0.155	0.485	0.071	0.575	0.226	0.462
2021	0.019	1.000	0.174	0.495	0.078	0.600	0.115	0.568	0.193	0.516
2022	0.000	0.000	0.035	0.698	0.000	0.000	0.035	0.698	0.035	0.698
2023	0.033	0.699	0.000	0.000	0.033	0.699	0.000	0.000	0.033	0.699
2024	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000

Table 16. Estimated annual abundance (millions of crab) of female PIBKC population components from the NMFS EBS trawl survey.

year	immature females		mature females		all females	
	est.	cv	est.	cv	est.	cv
1975	0.000	0.000	13.148	0.608	13.148	0.608
1976	7.369	0.966	0.769	0.513	8.139	0.910
1977	0.852	0.825	13.880	0.860	14.732	0.857
1978	0.061	1.000	5.927	0.662	5.987	0.656
1979	0.250	0.714	2.054	0.809	2.305	0.763
1980	0.781	0.774	182.903	0.977	183.684	0.976
1981	0.827	0.408	5.433	0.437	6.260	0.423
1982	0.876	0.514	7.837	0.648	8.713	0.626
1983	0.464	0.545	9.308	0.780	9.772	0.763
1984	0.465	0.516	2.769	0.380	3.235	0.366
1985	0.260	0.541	0.486	0.437	0.746	0.360
1986	0.037	0.698	2.102	0.898	2.139	0.882
1987	0.402	0.743	0.670	0.584	1.072	0.478
1988	0.898	0.869	0.465	0.479	1.363	0.642
1989	2.636	0.738	1.142	0.659	3.778	0.576
1990	2.177	0.910	2.046	0.547	4.223	0.555
1991	0.805	0.463	2.767	0.416	3.573	0.353
1992	1.797	0.927	2.150	0.494	3.947	0.521
1993	0.881	0.606	1.783	0.445	2.663	0.378
1994	0.145	0.574	5.047	0.443	5.192	0.437
1995	0.658	0.920	4.039	0.521	4.697	0.491
1996	0.276	0.418	5.046	0.484	5.322	0.463
1997	0.320	0.669	2.614	0.423	2.935	0.388
1998	0.500	0.431	1.830	0.443	2.330	0.365
1999	0.000	0.000	2.756	0.490	2.756	0.490
2000	0.000	0.000	1.363	0.463	1.363	0.463
2001	0.019	1.000	1.697	0.753	1.716	0.745
2002	0.019	1.000	1.222	0.794	1.241	0.782
2003	0.067	0.483	1.120	0.764	1.188	0.721
2004	0.098	0.634	0.070	0.603	0.168	0.510
2005	2.268	1.000	0.289	0.565	2.557	0.886
2006	0.113	0.548	0.430	0.766	0.543	0.617
2007	0.122	0.728	0.166	0.899	0.288	0.592
2008	0.342	0.898	0.437	0.658	0.779	0.748
2009	0.152	0.612	0.477	0.818	0.629	0.755
2010	0.166	0.558	0.249	0.691	0.415	0.622
2011	0.018	1.000	0.037	0.698	0.055	0.563
2012	0.035	1.000	0.312	0.764	0.347	0.695
2013	0.045	0.704	0.150	0.627	0.196	0.534
2014	0.028	1.000	0.074	0.604	0.102	0.507
2015	0.000	0.000	0.202	0.655	0.202	0.655
2016	0.095	0.515	0.359	0.520	0.454	0.504
2017	0.105	0.501	0.244	0.624	0.350	0.535

(continued)

year	immature females		mature females		all females	
	est.	cv	est.	cv	est.	cv
2018	0.020	1.000	0.114	0.614	0.134	0.537
2019	0.000	0.000	0.297	0.828	0.297	0.828
2021	0.000	0.000	0.229	0.671	0.229	0.671
2022	0.000	0.000	0.121	0.617	0.121	0.617
2023	0.000	0.000	0.117	1.000	0.117	1.000
2024	0.000	0.000	0.000	0.000	0.000	0.000

Table 17. Estimated annual biomass (1,000s t) of male PIBKC population components from the NMFS EBS trawl survey.

year	immature males		mature males		sublegal males		legal males		all males	
	est.	cv	est.	cv	est.	cv	est.	cv	est.	cv
1975	8.341	0.525	38.054	0.501	19.378	0.466	27.016	0.499	46.395	0.475
1976	4.129	0.944	14.059	0.451	5.539	0.811	12.649	0.468	18.188	0.452
1977	3.713	0.443	42.618	0.768	5.966	0.463	40.366	0.784	46.332	0.729
1978	2.765	0.509	17.370	0.558	6.618	0.412	13.517	0.642	20.135	0.506
1979	0.108	0.782	16.502	0.350	3.241	0.474	13.369	0.349	16.610	0.349
1980	2.084	0.492	23.553	0.430	4.958	0.464	20.679	0.446	25.637	0.417
1981	1.704	0.299	11.628	0.174	2.779	0.297	10.554	0.175	13.332	0.175
1982	1.152	0.232	7.389	0.187	1.647	0.217	6.893	0.192	8.541	0.175
1983	0.962	0.357	5.409	0.178	1.897	0.297	4.474	0.175	6.371	0.187
1984	0.130	0.362	2.216	0.229	0.521	0.268	1.824	0.247	2.345	0.222
1985	0.039	0.733	1.055	0.267	0.338	0.374	0.755	0.283	1.094	0.263
1986	0.004	1.000	1.505	0.303	0.035	0.897	1.473	0.307	1.508	0.302
1987	0.191	0.783	2.923	0.411	0.334	0.536	2.781	0.414	3.115	0.397
1988	0.170	0.707	0.842	0.529	0.170	0.707	0.842	0.529	1.012	0.457
1989	1.275	0.620	0.827	0.637	1.275	0.620	0.827	0.637	2.102	0.551
1990	2.004	0.661	3.078	0.600	3.567	0.665	1.514	0.515	5.082	0.610
1991	1.377	0.386	4.690	0.386	2.741	0.336	3.326	0.450	6.067	0.373
1992	1.801	0.512	4.391	0.423	3.157	0.446	3.035	0.446	6.192	0.432
1993	1.088	0.545	4.556	0.307	2.442	0.409	3.203	0.301	5.644	0.305
1994	0.619	0.388	3.410	0.345	1.224	0.350	2.806	0.351	4.029	0.343
1995	0.968	0.863	8.360	0.604	2.541	0.673	6.787	0.615	9.328	0.629
1996	0.745	0.605	4.641	0.269	1.512	0.524	3.873	0.265	5.386	0.279
1997	0.381	0.545	3.233	0.276	0.849	0.451	2.765	0.271	3.614	0.294
1998	0.692	0.413	2.798	0.249	0.980	0.354	2.510	0.255	3.490	0.252
1999	0.161	0.402	1.729	0.337	0.464	0.414	1.426	0.347	1.890	0.333
2000	0.113	0.679	2.091	0.296	0.459	0.373	1.746	0.305	2.205	0.304
2001	0.087	0.764	1.599	0.735	0.225	0.628	1.461	0.759	1.686	0.733
2002	0.000	0.000	0.680	0.506	0.033	1.000	0.647	0.525	0.680	0.506
2003	0.019	0.984	0.702	0.400	0.050	0.723	0.671	0.411	0.721	0.390
2004	0.036	0.649	0.107	0.583	0.094	0.487	0.048	1.000	0.143	0.455
2005	0.326	0.942	0.344	0.710	0.326	0.942	0.344	0.710	0.670	0.589
2006	0.087	0.585	0.166	0.603	0.114	0.616	0.139	0.699	0.253	0.462
2007	0.197	0.737	0.306	0.798	0.298	0.632	0.206	0.734	0.503	0.661
2008	0.212	0.952	0.046	1.000	0.212	0.952	0.046	1.000	0.258	0.797
2009	0.254	0.680	0.497	0.713	0.565	0.740	0.187	0.604	0.751	0.698
2010	0.092	0.853	0.303	0.461	0.205	0.702	0.190	0.483	0.395	0.522
2011	0.000	0.000	0.461	0.843	0.062	0.705	0.399	0.886	0.461	0.843
2012	0.165	1.000	0.644	0.735	0.350	1.000	0.459	0.643	0.809	0.786
2013	0.015	1.000	0.250	0.797	0.075	0.824	0.190	0.752	0.265	0.754
2014	0.083	0.623	0.233	0.699	0.083	0.623	0.233	0.699	0.317	0.567
2015	0.082	0.747	0.622	0.394	0.275	0.494	0.428	0.458	0.703	0.395
2016	0.071	0.486	0.130	0.613	0.133	0.495	0.068	1.000	0.201	0.515
2017	0.046	0.767	0.255	0.514	0.076	0.599	0.224	0.573	0.300	0.470

(continued)

year	immature males		mature males		sublegal males		legal males		all males	
	est.	cv	est.	cv	est.	cv	est.	cv	est.	cv
2018	0.096	0.540	0.154	0.571	0.096	0.540	0.154	0.571	0.249	0.522
2019	0.115	0.542	0.206	0.604	0.115	0.542	0.206	0.604	0.321	0.504
2021	0.015	1.000	0.405	0.503	0.122	0.653	0.298	0.576	0.420	0.512
2022	0.000	0.000	0.112	0.702	0.000	0.000	0.112	0.702	0.112	0.702
2023	0.024	1.000	0.000	0.000	0.024	1.000	0.000	0.000	0.024	1.000
2024	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000

Table 18. Estimated annual biomass (1,000s t) of female PIBKC population components from the NMFS EBS trawl survey.

year	immature females		mature females		all females	
	est.	cv	est.	cv	est.	cv
1975	0.000	0.000	12.442	0.636	12.442	0.636
1976	4.968	0.972	0.824	0.532	5.792	0.891
1977	0.419	0.829	13.154	0.875	13.572	0.874
1978	0.076	1.000	6.416	0.725	6.492	0.717
1979	0.161	0.725	1.929	0.790	2.090	0.756
1980	0.699	0.865	211.604	0.984	212.303	0.983
1981	0.497	0.413	5.987	0.469	6.484	0.458
1982	0.553	0.572	8.824	0.678	9.377	0.669
1983	0.258	0.607	9.990	0.791	10.248	0.781
1984	0.015	0.688	3.070	0.381	3.085	0.380
1985	0.005	0.457	0.520	0.448	0.525	0.445
1986	0.011	0.727	2.420	0.901	2.431	0.896
1987	0.119	0.855	0.795	0.583	0.913	0.526
1988	0.190	0.788	0.528	0.491	0.718	0.473
1989	0.801	0.666	0.945	0.581	1.746	0.497
1990	1.118	0.928	1.810	0.508	2.929	0.491
1991	0.343	0.475	2.433	0.414	2.776	0.376
1992	0.802	0.961	1.848	0.480	2.649	0.463
1993	0.444	0.624	1.647	0.461	2.092	0.399
1994	0.087	0.570	4.806	0.447	4.893	0.443
1995	0.331	0.904	3.948	0.519	4.279	0.496
1996	0.177	0.415	5.408	0.502	5.585	0.491
1997	0.194	0.659	2.835	0.429	3.028	0.407
1998	0.267	0.425	1.914	0.441	2.182	0.392
1999	0.000	0.000	2.868	0.467	2.868	0.467
2000	0.000	0.000	1.462	0.460	1.462	0.460
2001	0.000	1.000	1.816	0.722	1.817	0.722
2002	0.000	1.000	1.401	0.776	1.401	0.775
2003	0.021	0.667	1.286	0.745	1.307	0.734
2004	0.025	0.821	0.098	0.597	0.123	0.504
2005	0.477	1.000	0.370	0.570	0.847	0.606
2006	0.038	0.602	0.538	0.760	0.576	0.712
2007	0.059	0.792	0.223	0.876	0.282	0.707
2008	0.222	0.901	0.450	0.635	0.672	0.705
2009	0.080	0.660	0.545	0.849	0.625	0.818
2010	0.084	0.578	0.310	0.660	0.394	0.634
2011	0.003	1.000	0.034	0.725	0.037	0.674
2012	0.009	1.000	0.229	0.660	0.237	0.637
2013	0.012	0.722	0.154	0.700	0.166	0.654
2014	0.016	1.000	0.091	0.605	0.108	0.529
2015	0.000	0.000	0.160	0.662	0.160	0.662
2016	0.050	0.490	0.354	0.493	0.405	0.478
2017	0.055	0.501	0.206	0.591	0.262	0.533

(continued)

year	immature females		mature females		all females	
	est.	cv	est.	cv	est.	cv
2018	0.013	1.000	0.108	0.725	0.121	0.654
2019	0.000	0.000	0.412	0.859	0.412	0.859
2021	0.000	0.000	0.262	0.632	0.262	0.632
2022	0.000	0.000	0.146	0.663	0.146	0.663
2023	0.000	0.000	0.119	1.000	0.119	1.000
2024	0.000	0.000	0.000	0.000	0.000	0.000

Figures

List of Figures

1	NMFS EBS Shelf Survey stations in the Pribilof District (large dots), the survey station grid (thin black lines), and the Pribilof Islands Habitat Conservation Zone (orange outline).	21
2	NMFS survey abundance time series for male PIBKC, by maturity category.	22
3	NMFS survey abundance time series for male PIBKC, by fishery category.	23
4	NMFS survey abundance time series for male PIBKC, by population category, from 2010.	24
5	NMFS survey abundance time series for male PIBKC, by fishery category, from 2010.	25
6	NMFS survey abundance time series for female PIBKC, by population category. The values for mature and all females for 1980 are off-scale to better show details of remaining values.	26
7	NMFS survey abundance time series for female PIBKC, by population category, from 2010.	27
8	NMFS survey biomass time series for male PIBKC, by maturity category.	28
9	NMFS survey biomass time series for male PIBKC, by fishery category.	29
10	NMFS survey biomass time series for male PIBKC, by maturity category, from 2010.	30
11	NMFS survey biomass time series for male PIBKC, by fishery category, from 2010.	31
12	NMFS survey biomass time series for female PIBKC, by population category. The values for mature and all females for 1980 are off-scale to better show details of remaining values.	32
13	NMFS survey biomass time series for female PIBKC, by population category, from 2010.	33
14	Annual size compositions for PIBKC in the NMFS EBS trawl survey, by sex, over the entire survey period. The survey was not conducted in 2020.	34
15	Annual size compositions for PIBKC in the NMFS EBS trawl survey, by sex, over the entire survey period, except that females in 1980 have been removed to show detail. The survey was not conducted in 2020.	35
16	Annual size compositions for PIBKC in the NMFS EBS trawl survey, by sex, since 2006. The survey was not conducted in 2020.	36
17	Decadal-average abundance CPUE (number/sq-nmi) by for male PIBKC in the NMFS EBS trawl survey.	37
18	Decadal-average abundance CPUE (number/sq-nmi) by for female PIBKC in the NMFS EBS trawl survey.	38
19	Decadal-average biomass CPUE (t/sq-nmi) by for male PIBKC in the NMFS EBS trawl survey.	39
20	Decadal-average biomass CPUE (t/sq-nmi) by for female PIBKC in the NMFS EBS trawl survey.	40

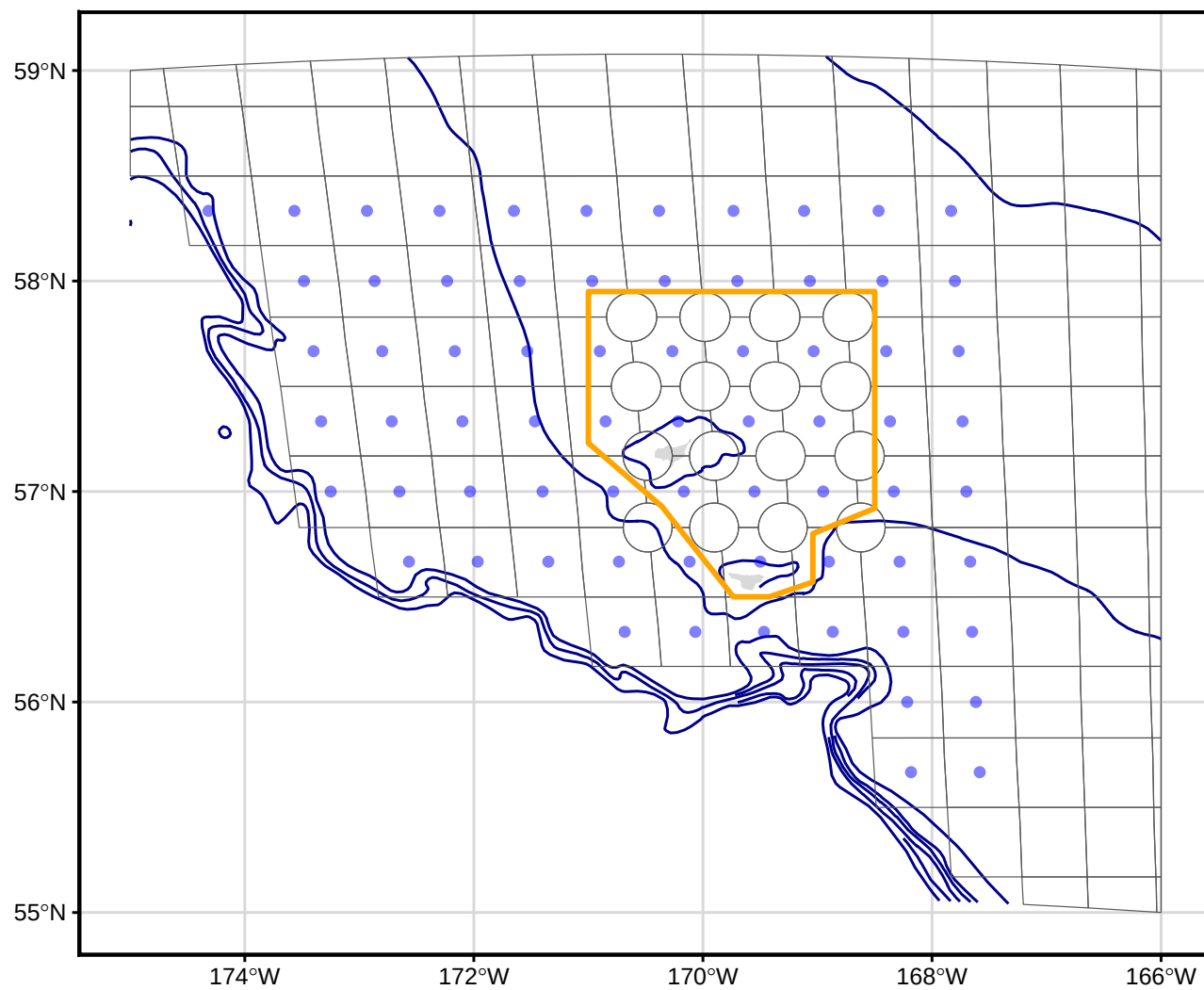


Figure 1. NMFS EBS Shelf Survey stations in the Pribilof District (large dots), the survey station grid (thin black lines), and the Pribilof Islands Habitat Conservation Zone (orange outline).

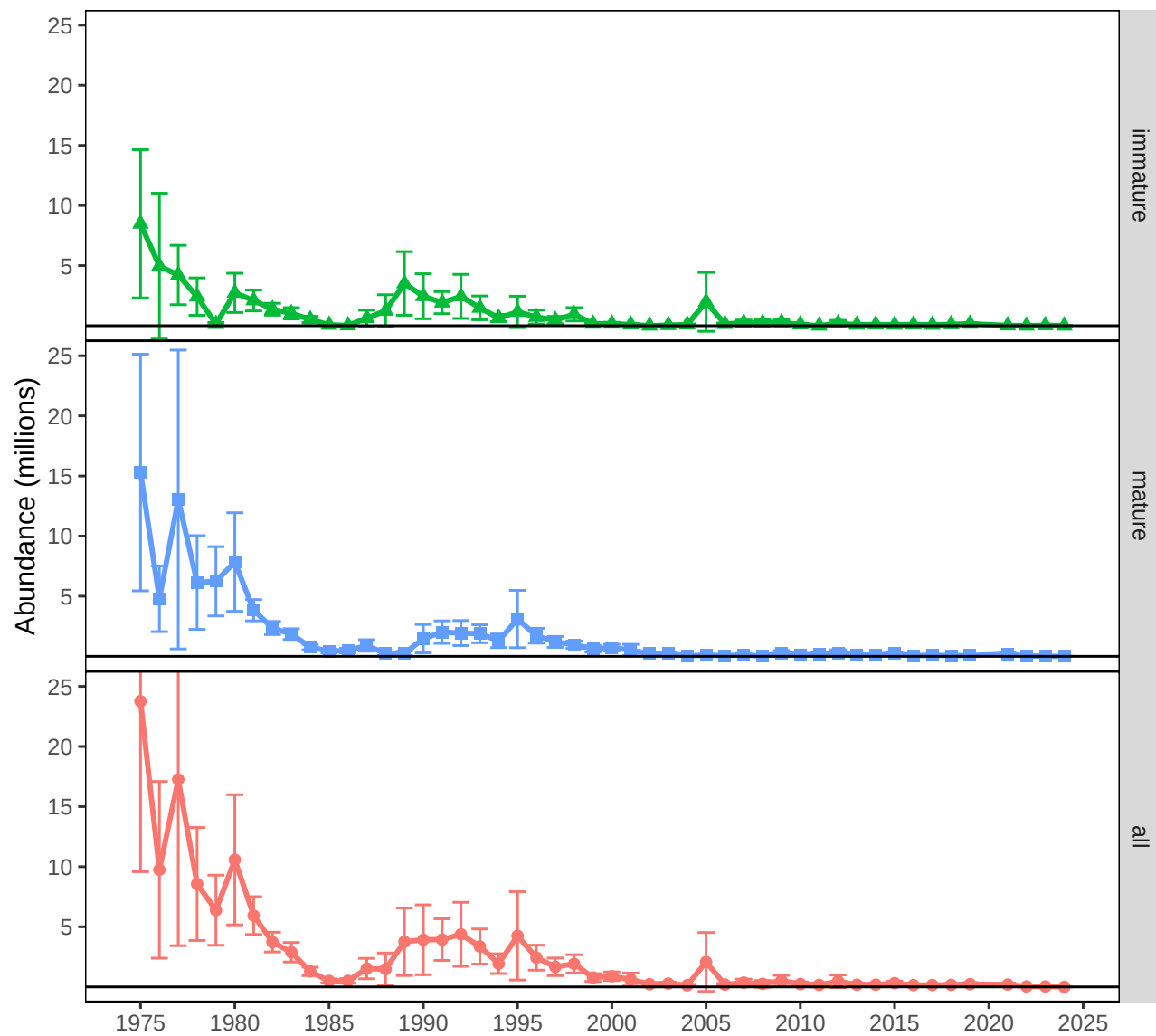


Figure 2. NMFS survey abundance time series for male PIBKC, by maturity category.

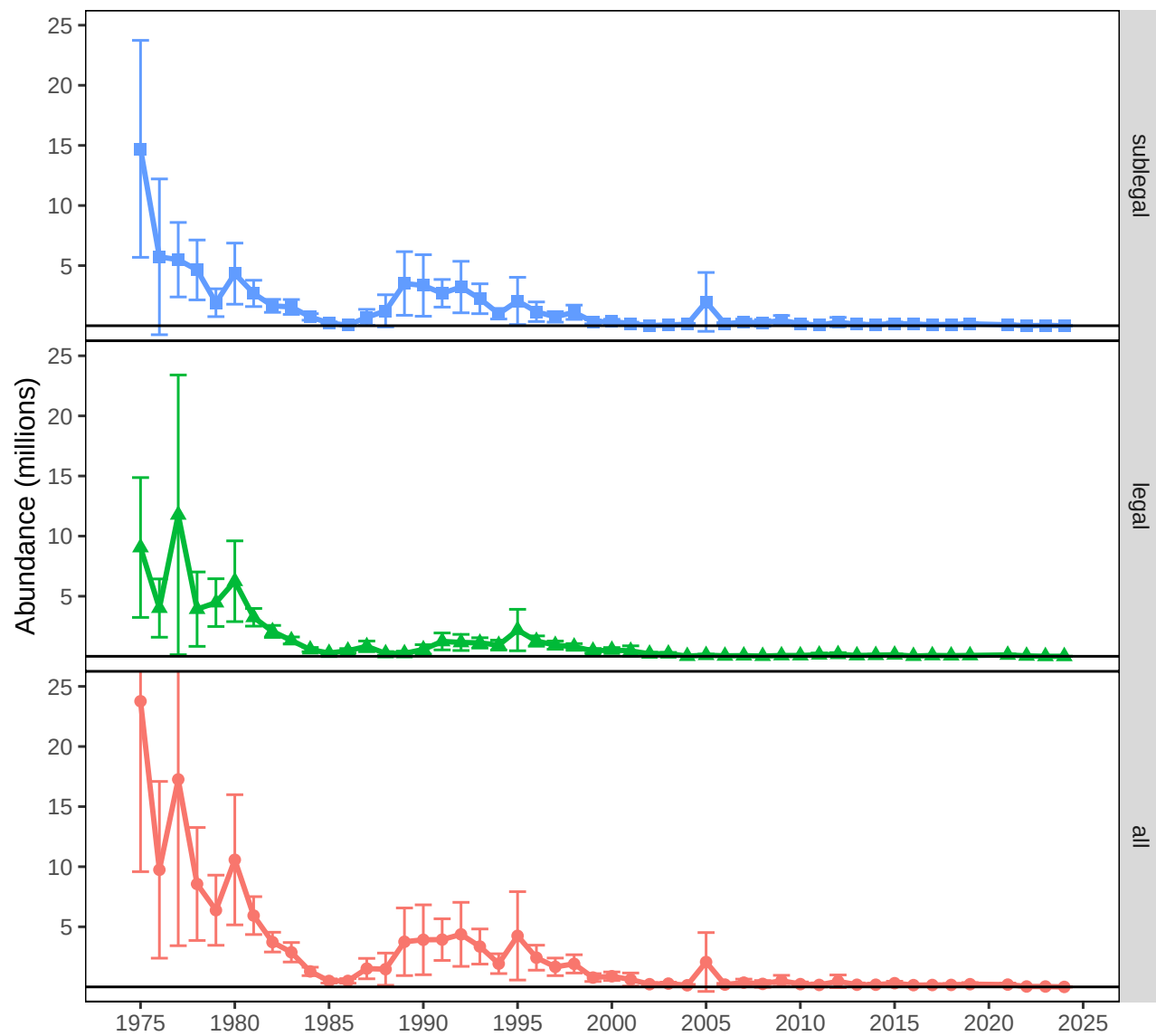


Figure 3. NMFS survey abundance time series for male PIBKC, by fishery category.

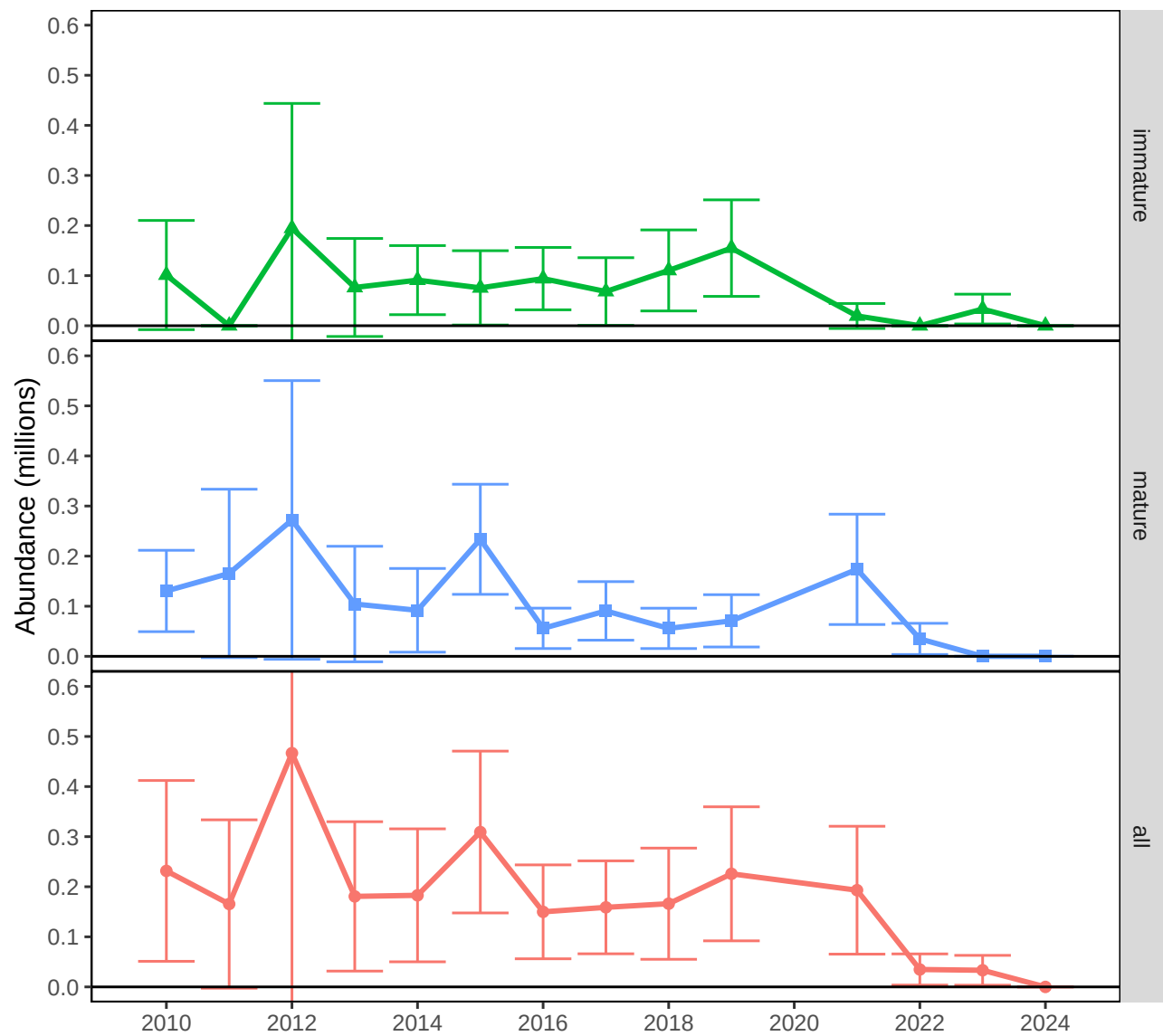


Figure 4. NMFS survey abundance time series for male PIBKC, by population category, from 2010.

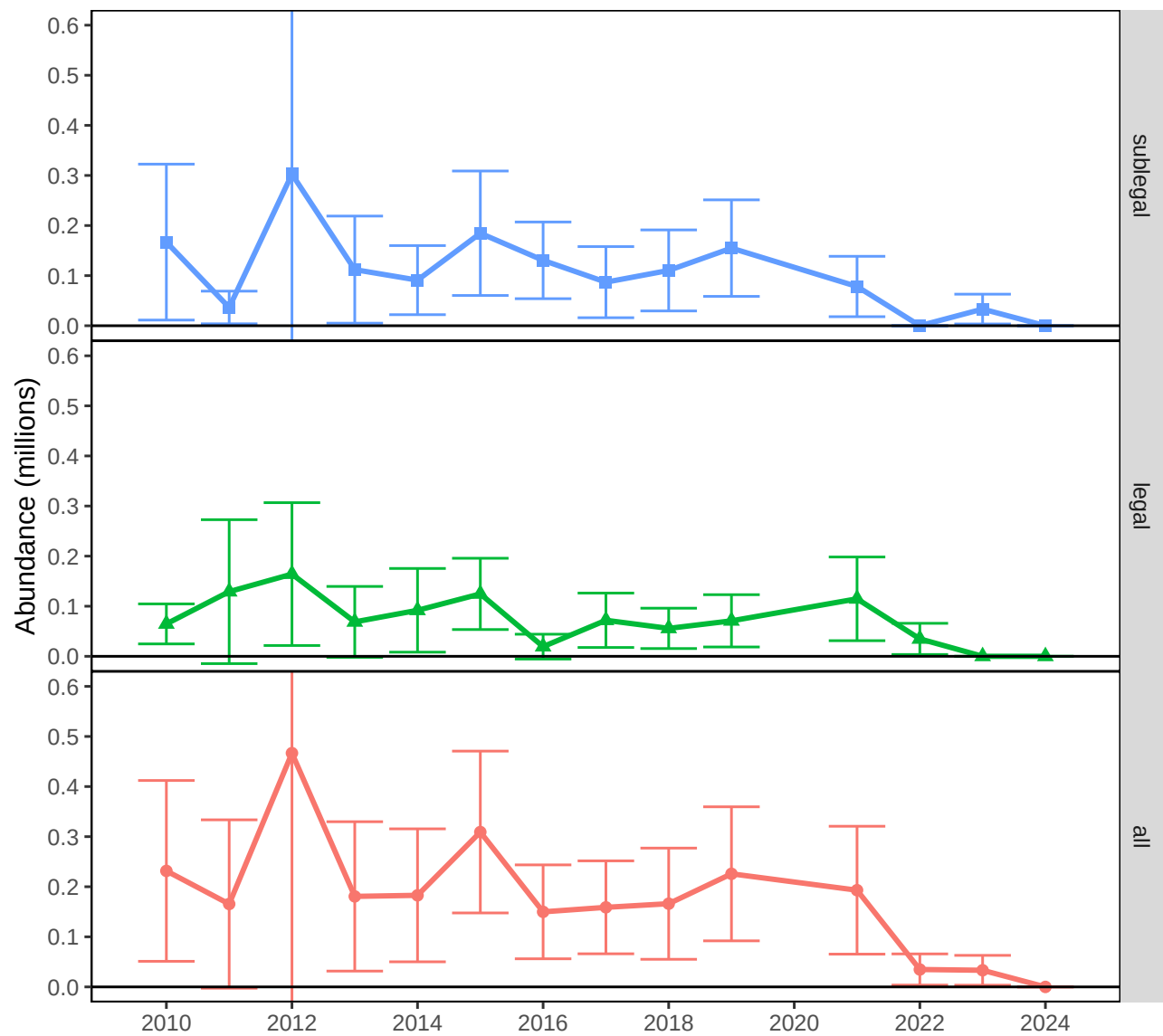


Figure 5. NMFS survey abundance time series for male PIBKC, by fishery category, from 2010.

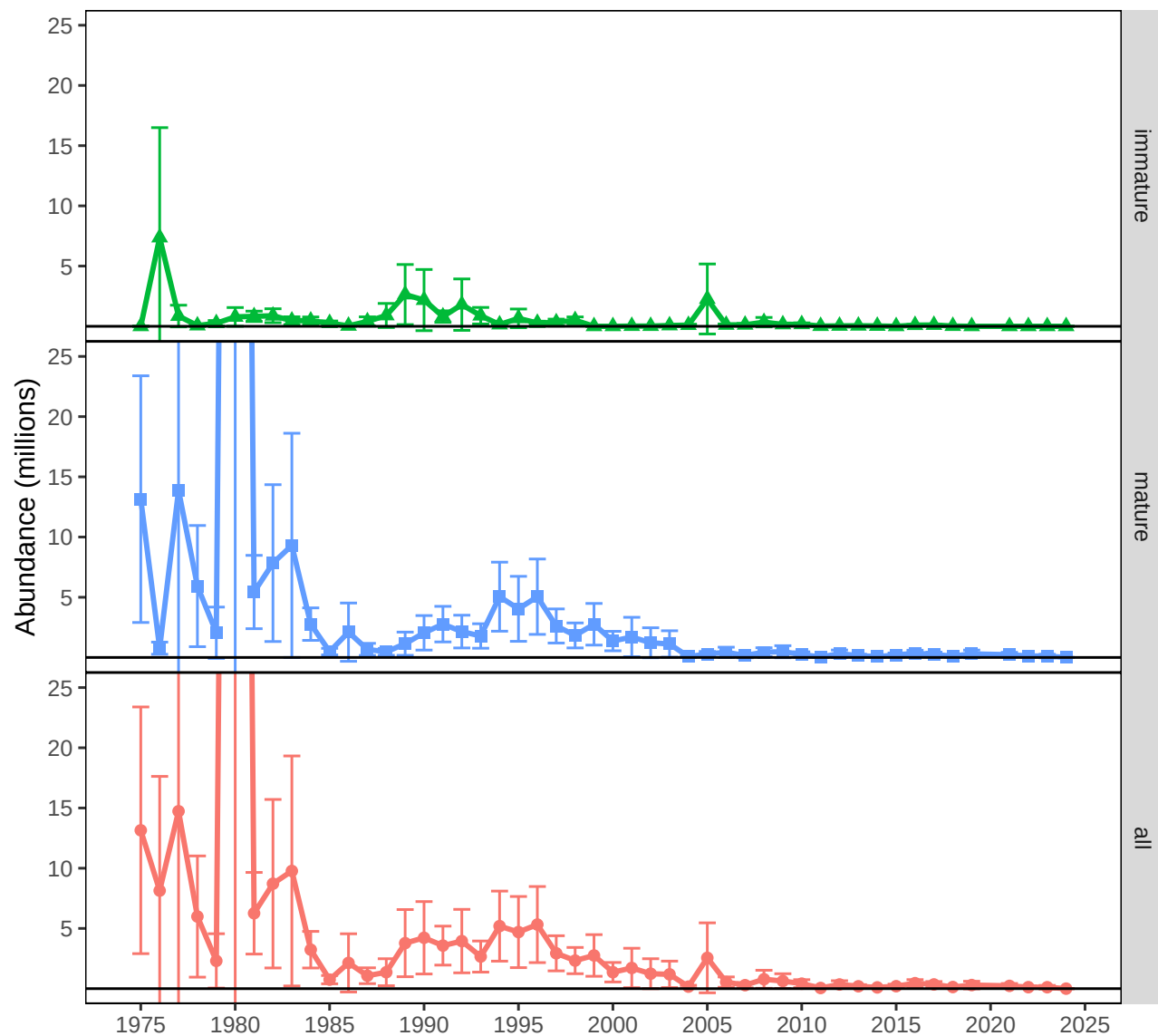


Figure 6. NMFS survey abundance time series for female PIBKC, by population category. The values for mature and all females for 1980 are off-scale to better show details of remaining values.

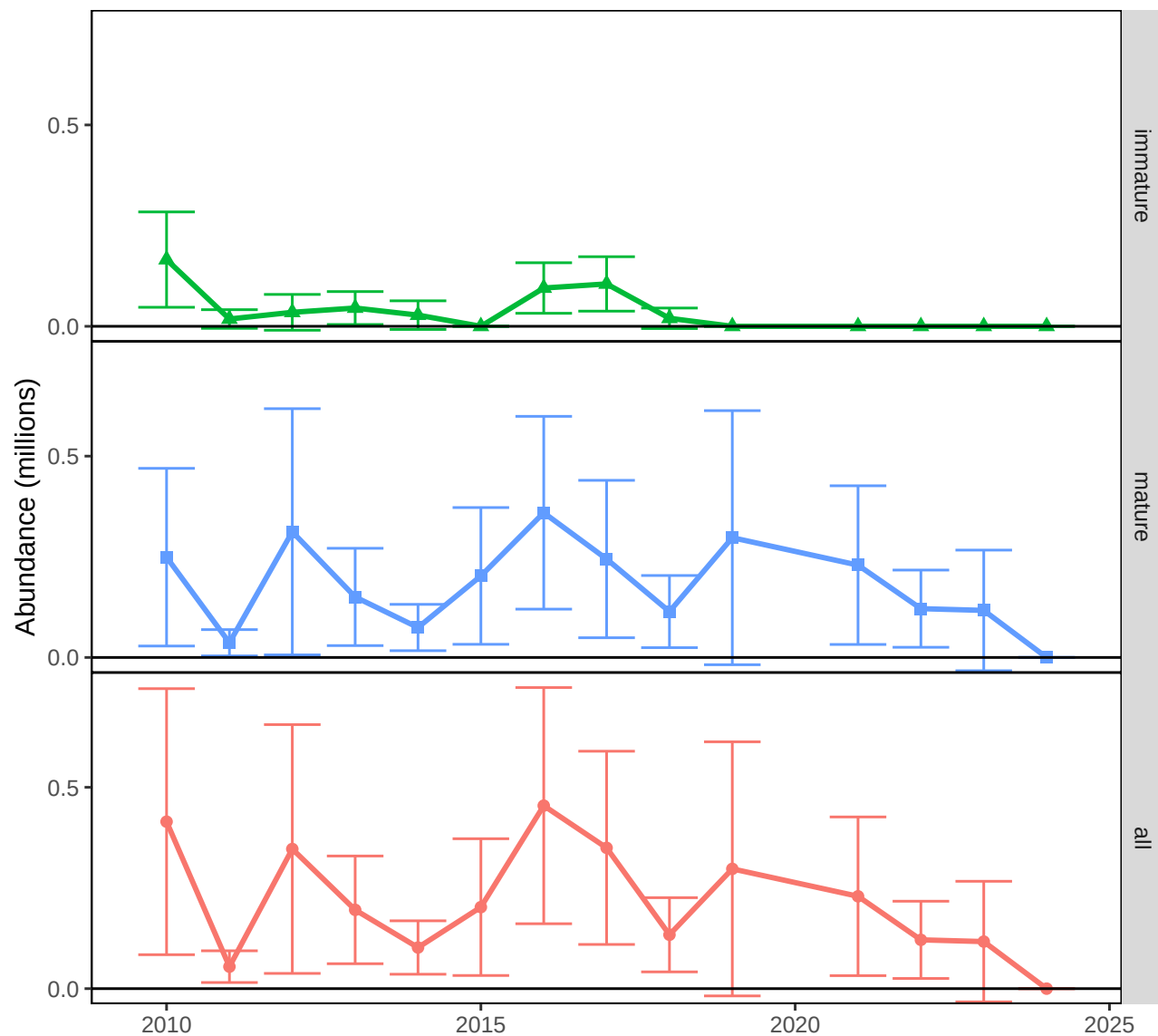


Figure 7. NMFS survey abundance time series for female PIBKC, by population category, from 2010.

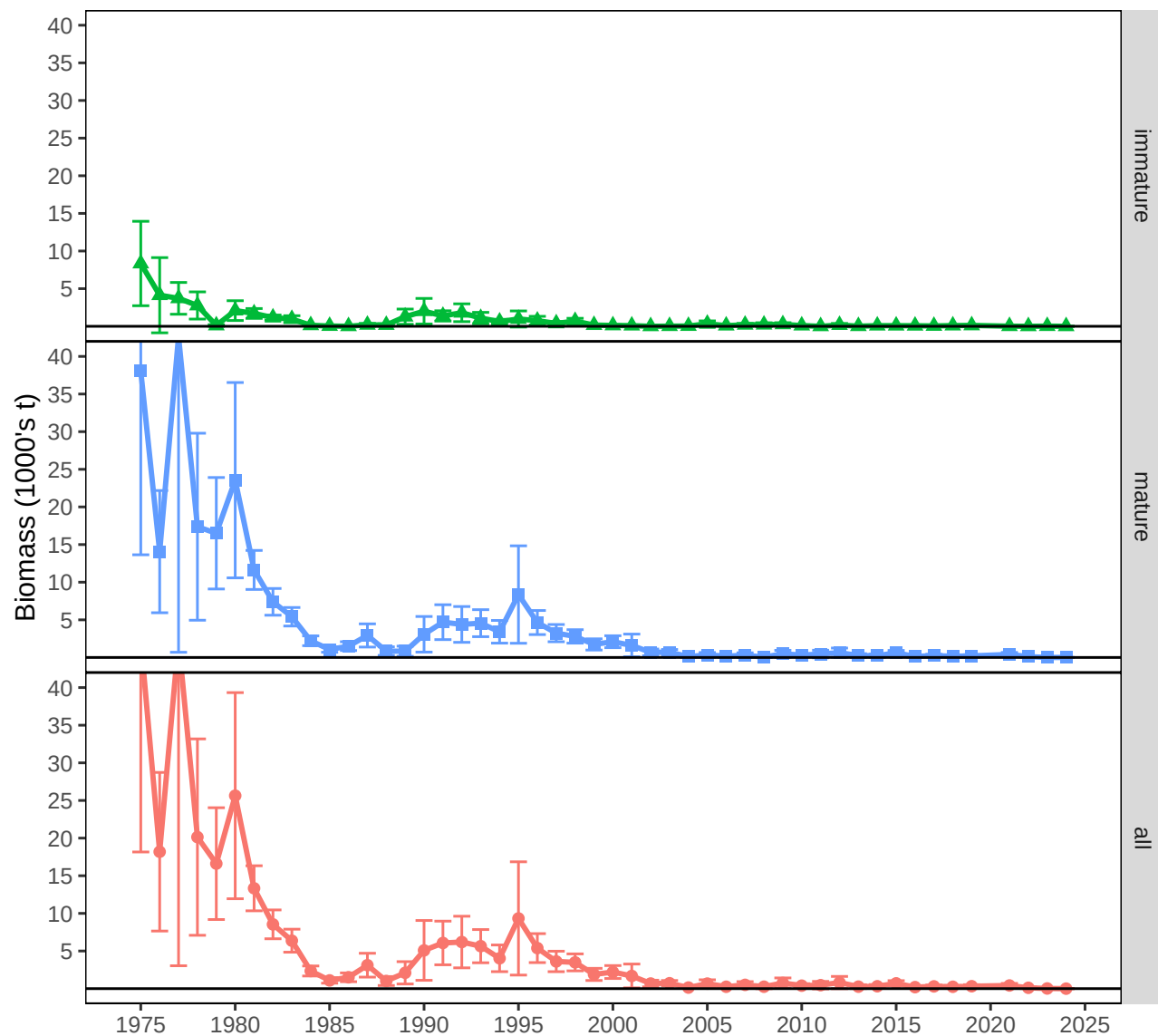


Figure 8. NMFS survey biomass time series for male PIBKC, by maturity category.

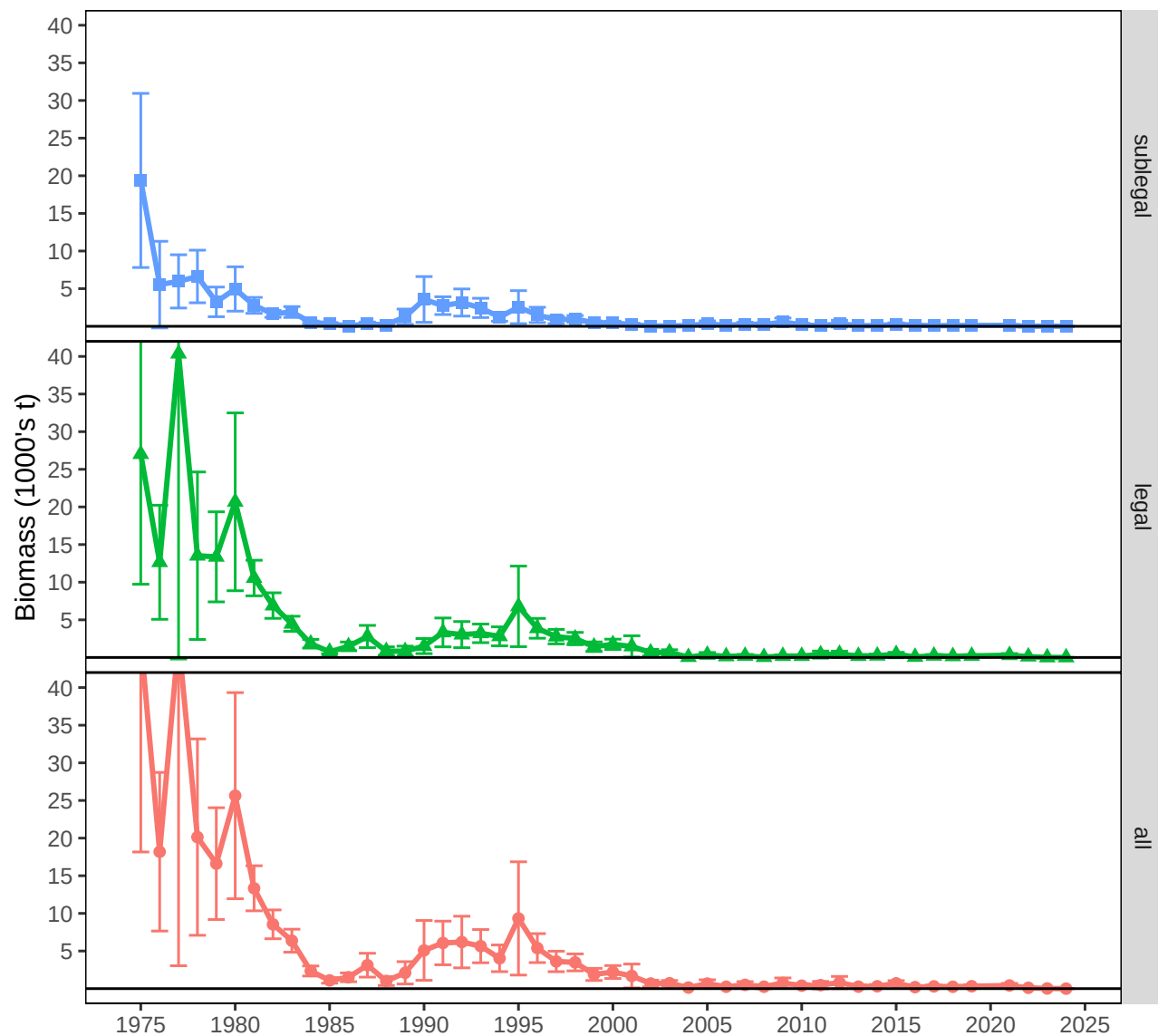


Figure 9. NMFS survey biomass time series for male PIBKC, by fishery category.

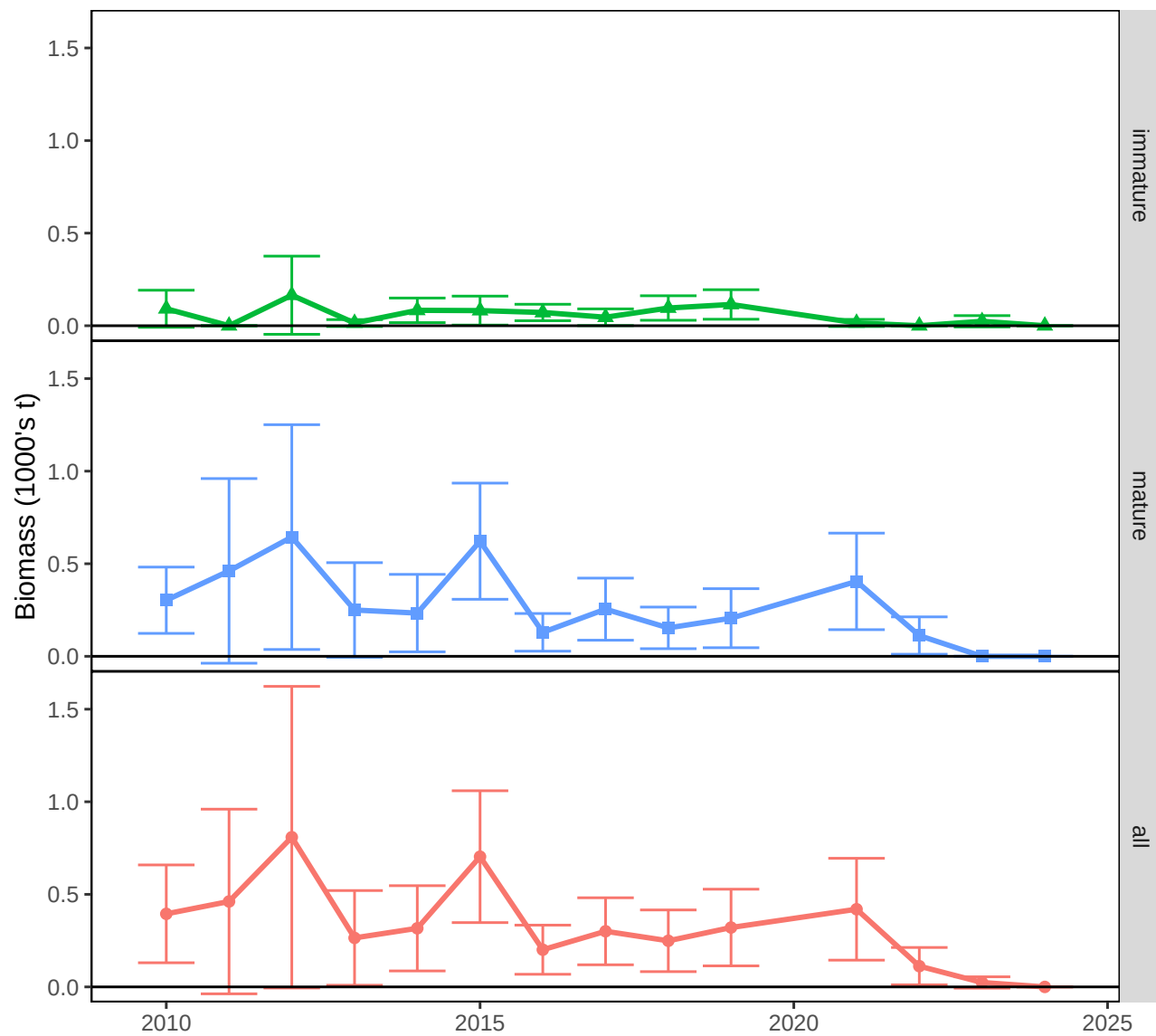


Figure 10. NMFS survey biomass time series for male PIBKC, by maturity category, from 2010.

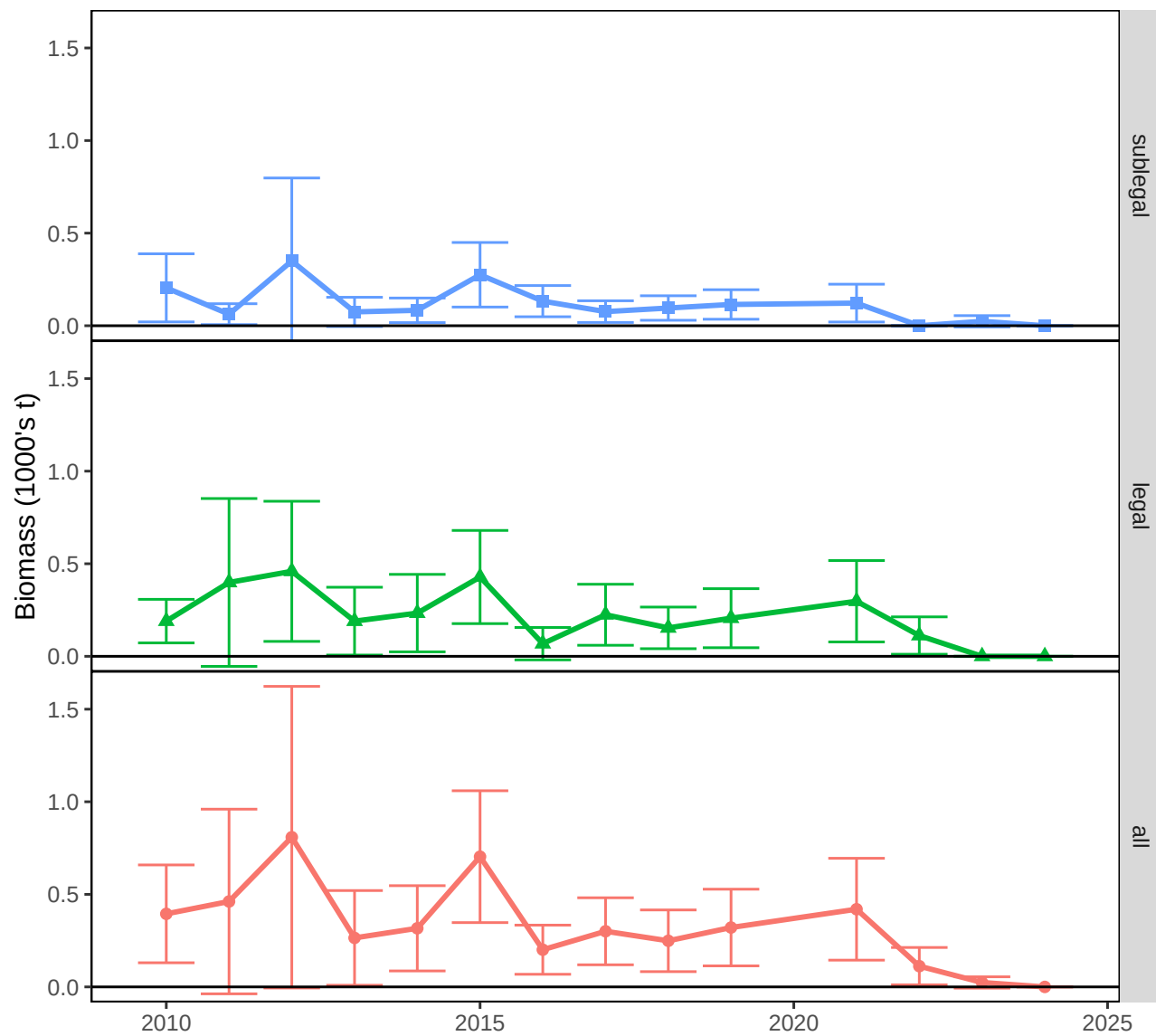


Figure 11. NMFS survey biomass time series for male PIBKC, by fishery category, from 2010.

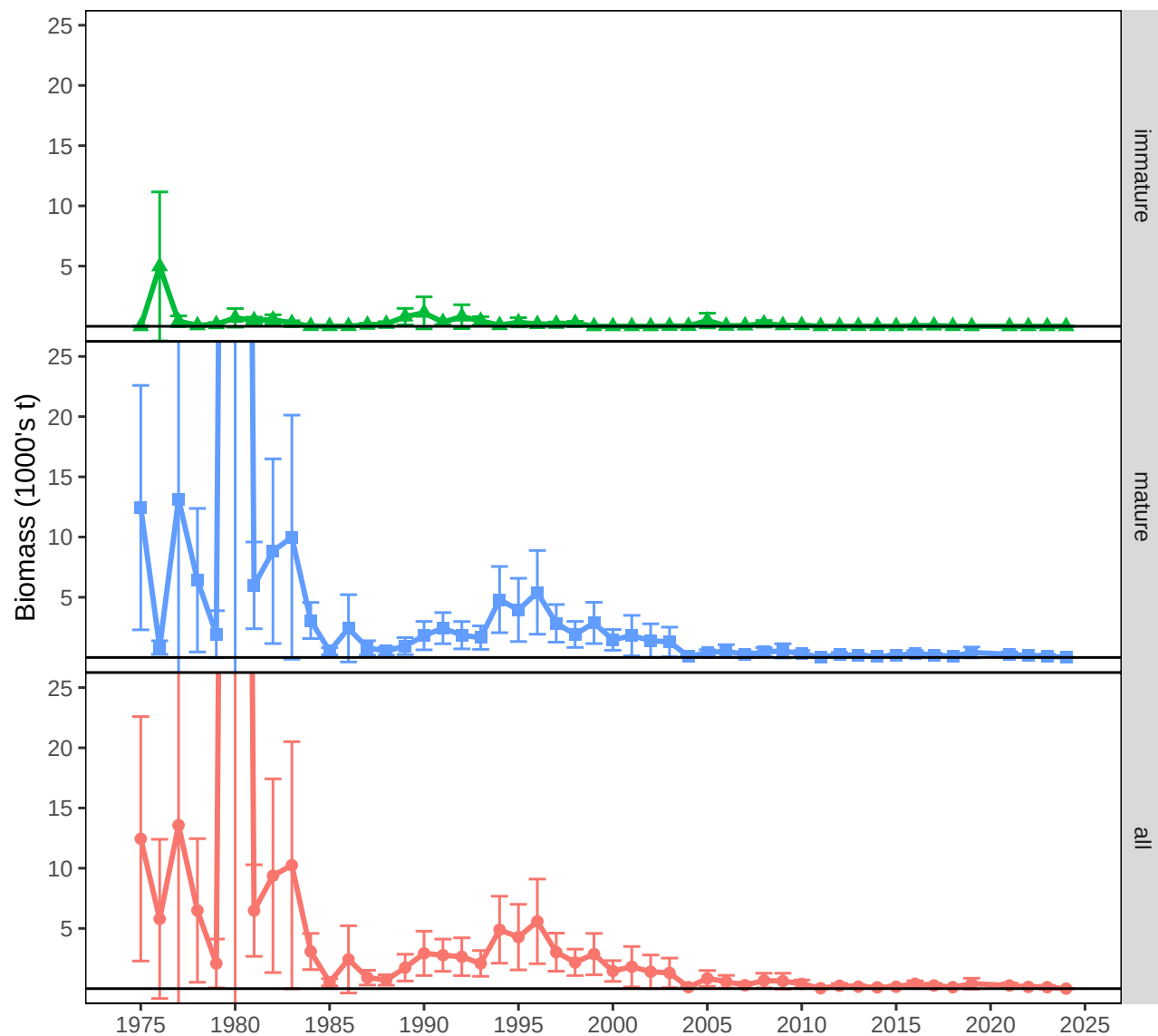


Figure 12. NMFS survey biomass time series for female PIBKC, by population category. The values for mature and all females for 1980 are off-scale to better show details of remaining values.

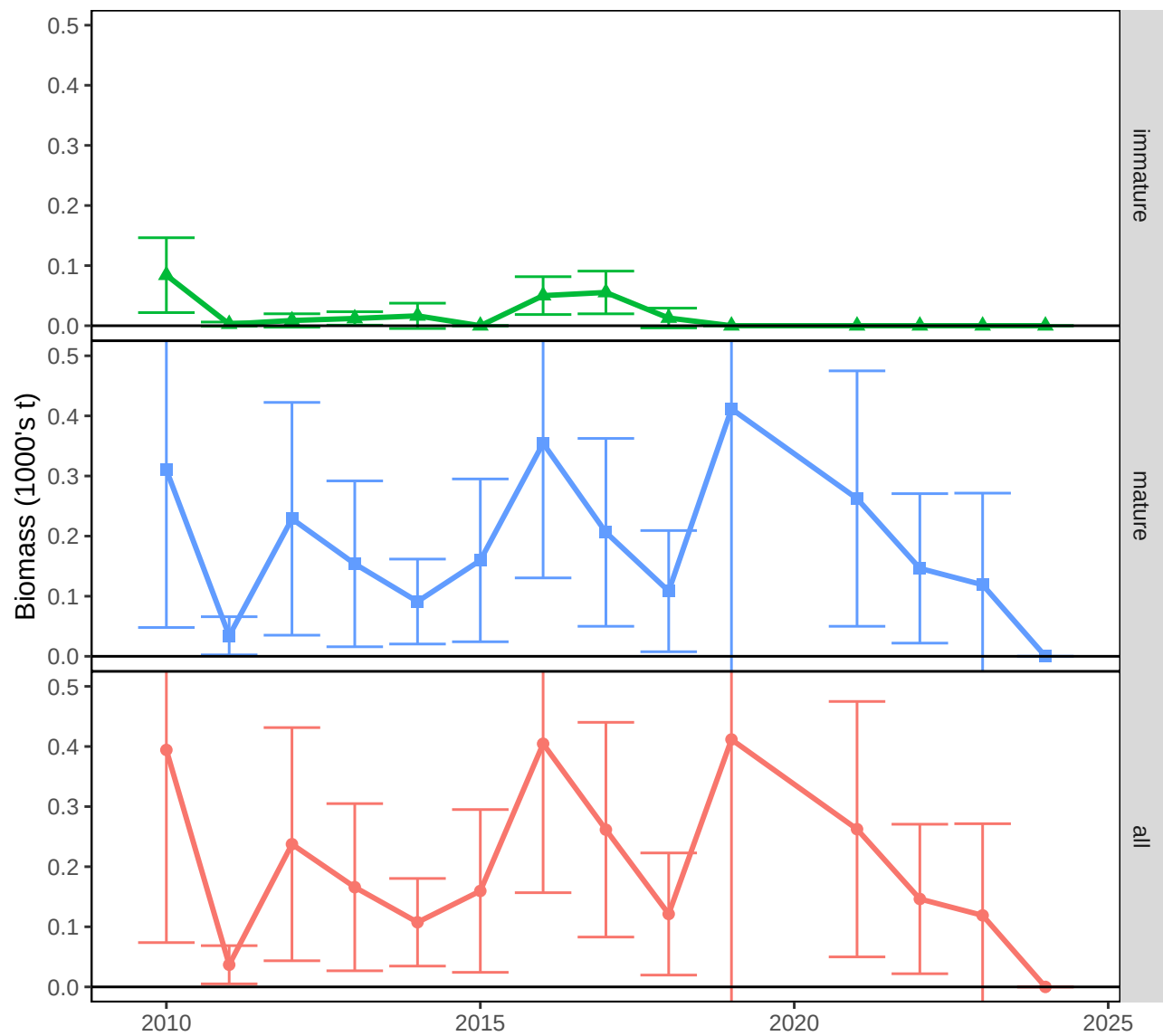


Figure 13. NMFS survey biomass time series for female PIBKC, by population category, from 2010.

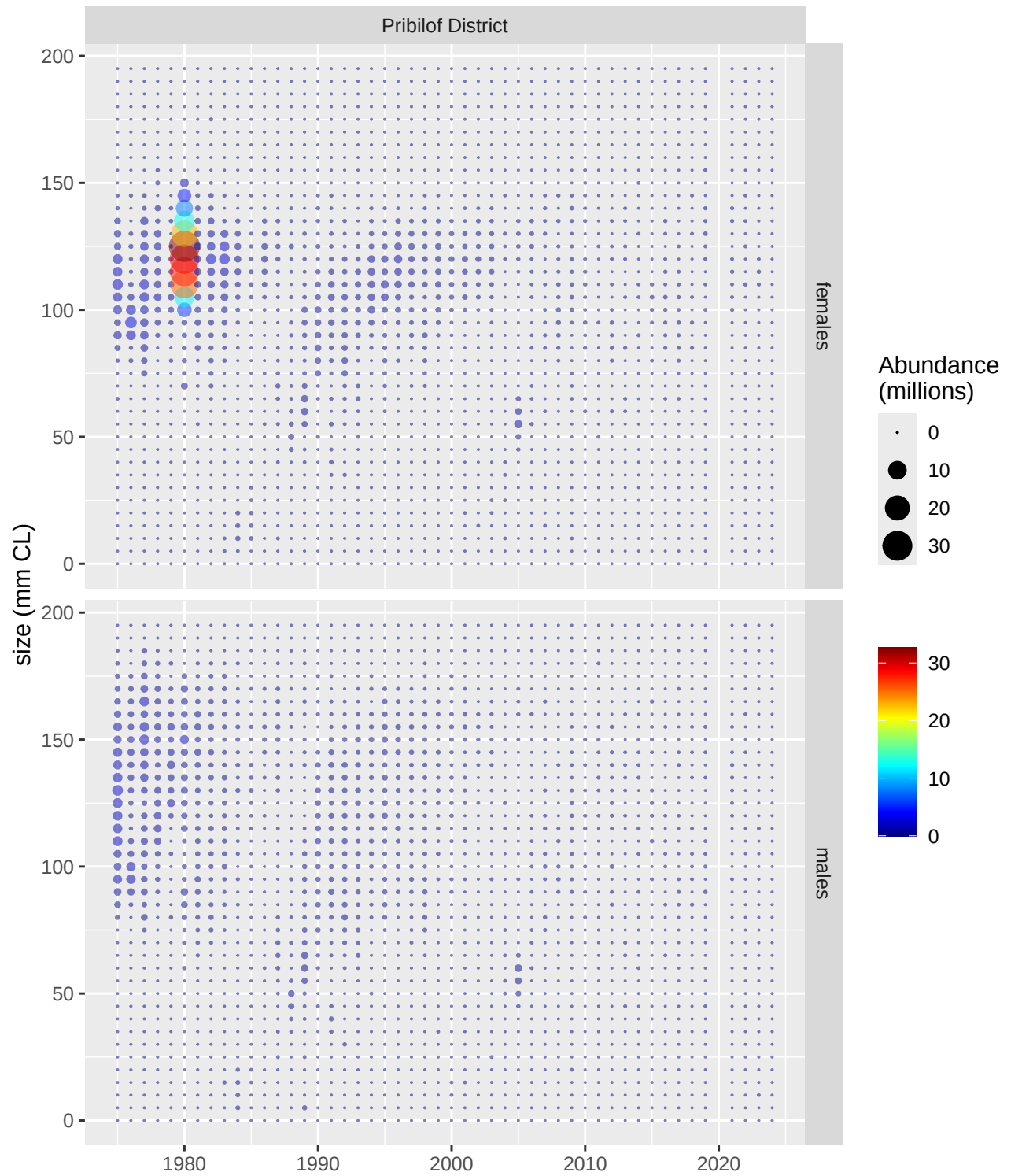


Figure 14. Annual size compositions for PIBKC in the NMFS EBS trawl survey, by sex, over the entire survey period. The survey was not conducted in 2020.

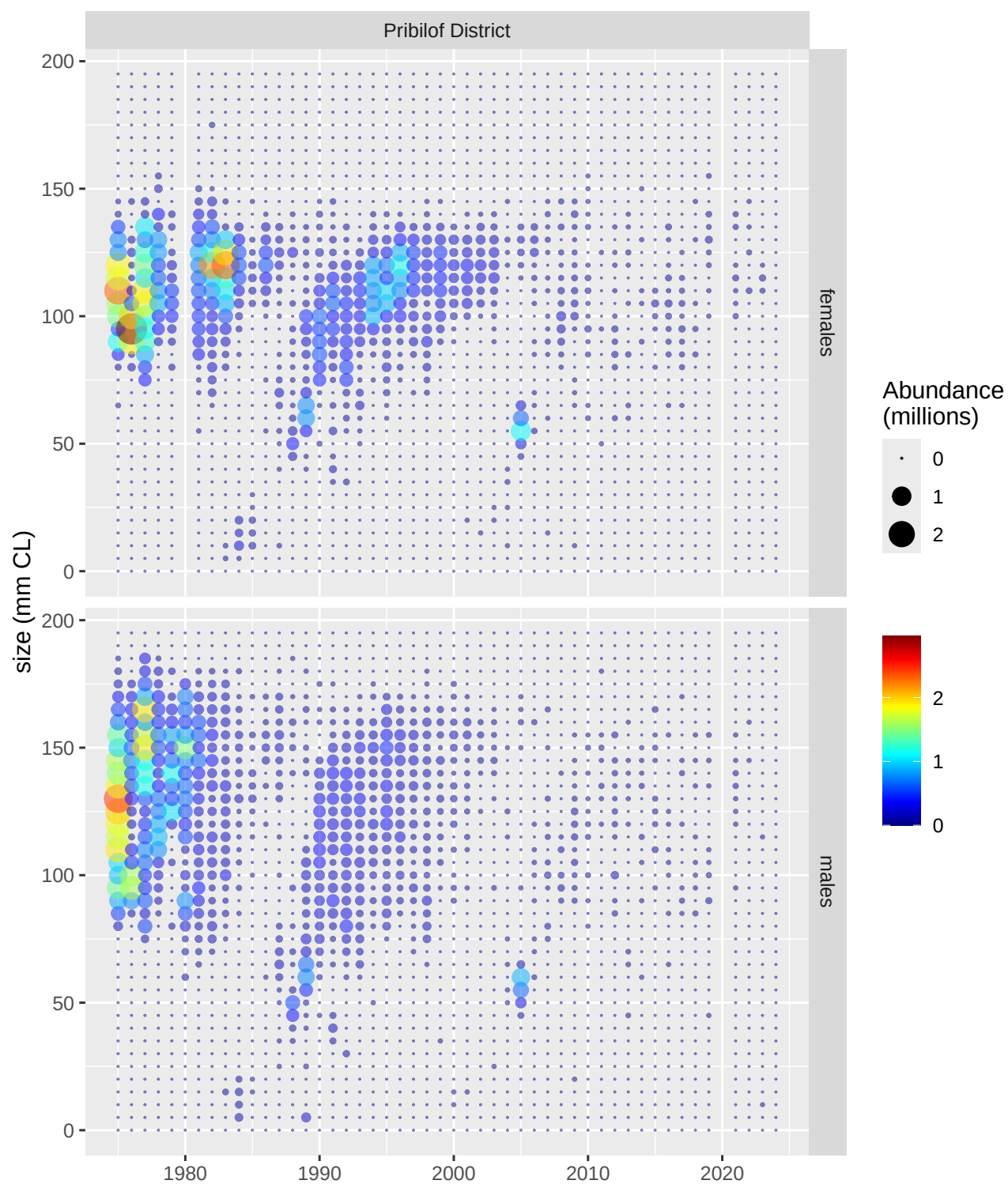


Figure 15. Annual size compositions for PIBKC in the NMFS EBS trawl survey, by sex, over the entire survey period, except that females in 1980 have been removed to show detail. The survey was not conducted in 2020.

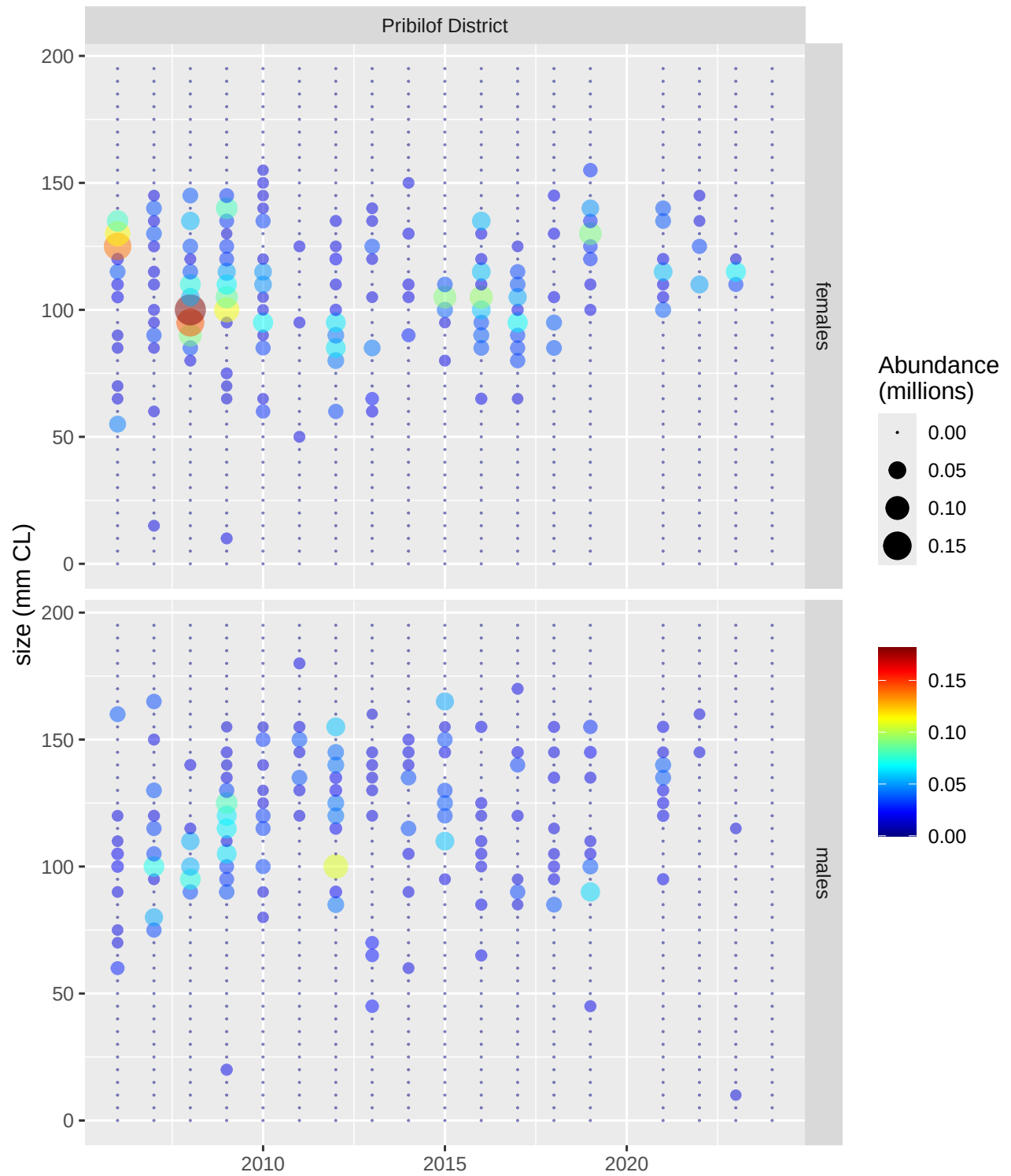


Figure 16. Annual size compositions for PIBKC in the NMFS EBS trawl survey, by sex, since 2006. The survey was not conducted in 2020.

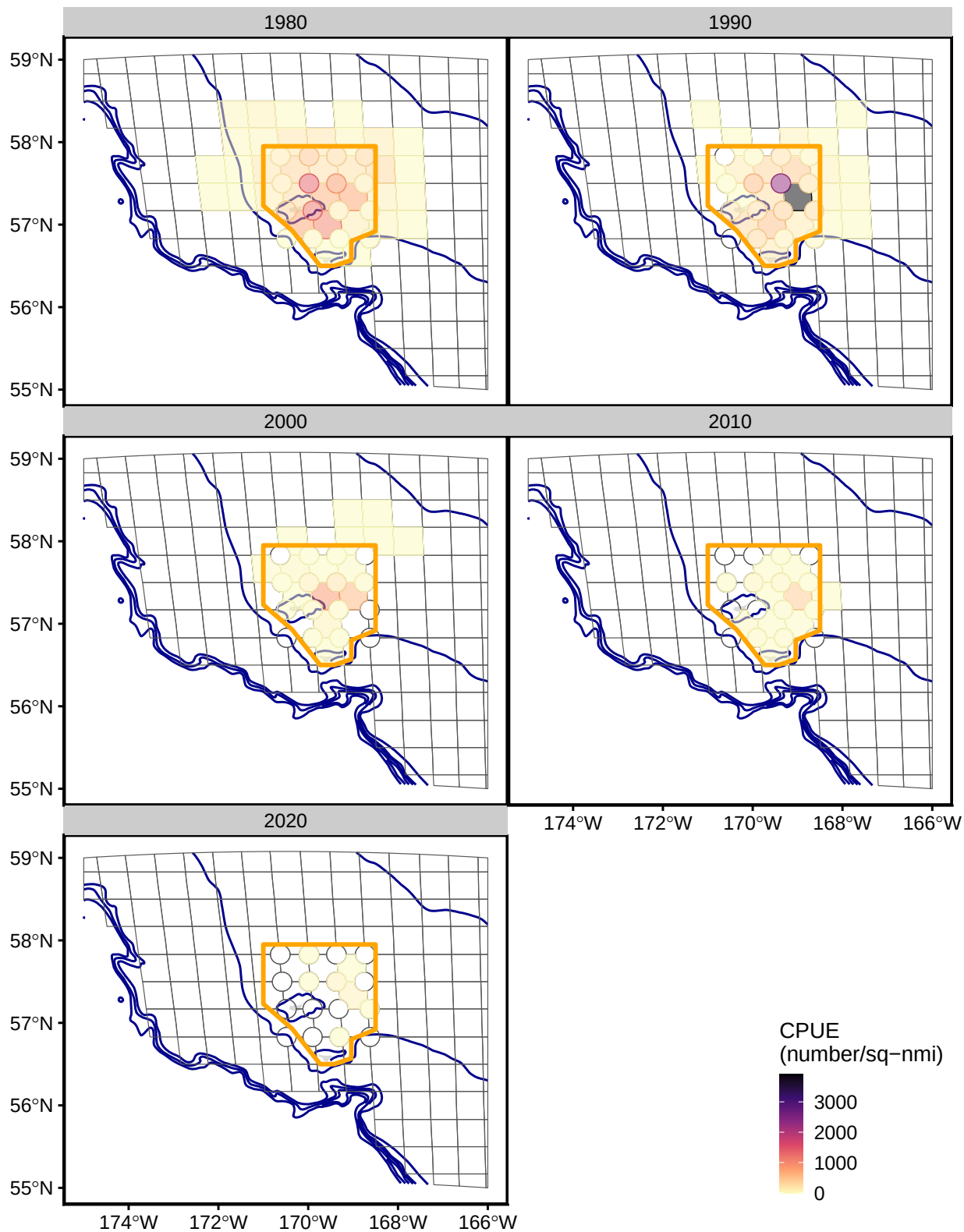


Figure 17. Decadal-average abundance CPUE (number/sq-nmi) by for male PIBKC in the NMFS EBS trawl survey.

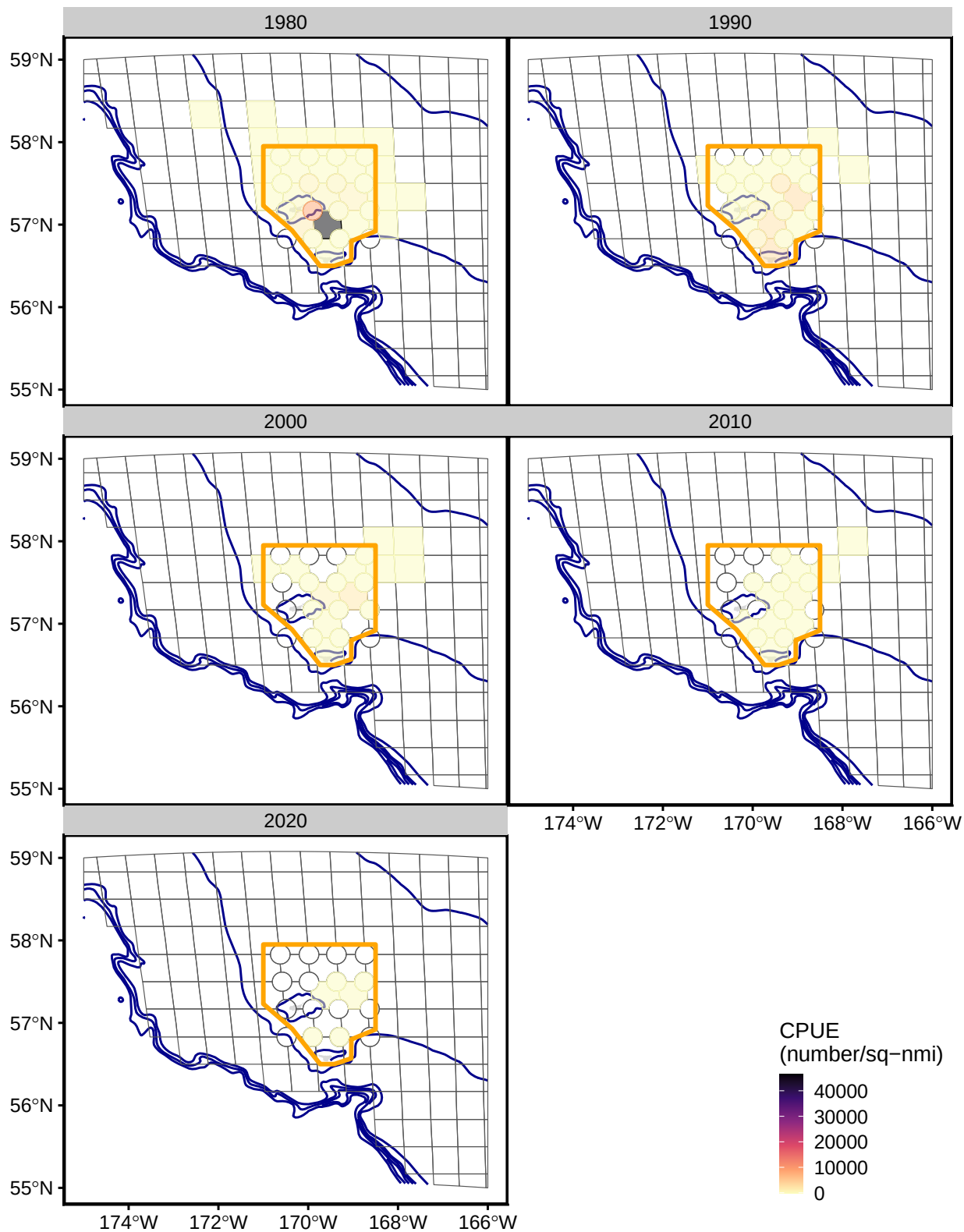


Figure 18. Decadal-average abundance CPUE (number/sq-nmi) by for female PIBKC in the NMFS EBS trawl survey.

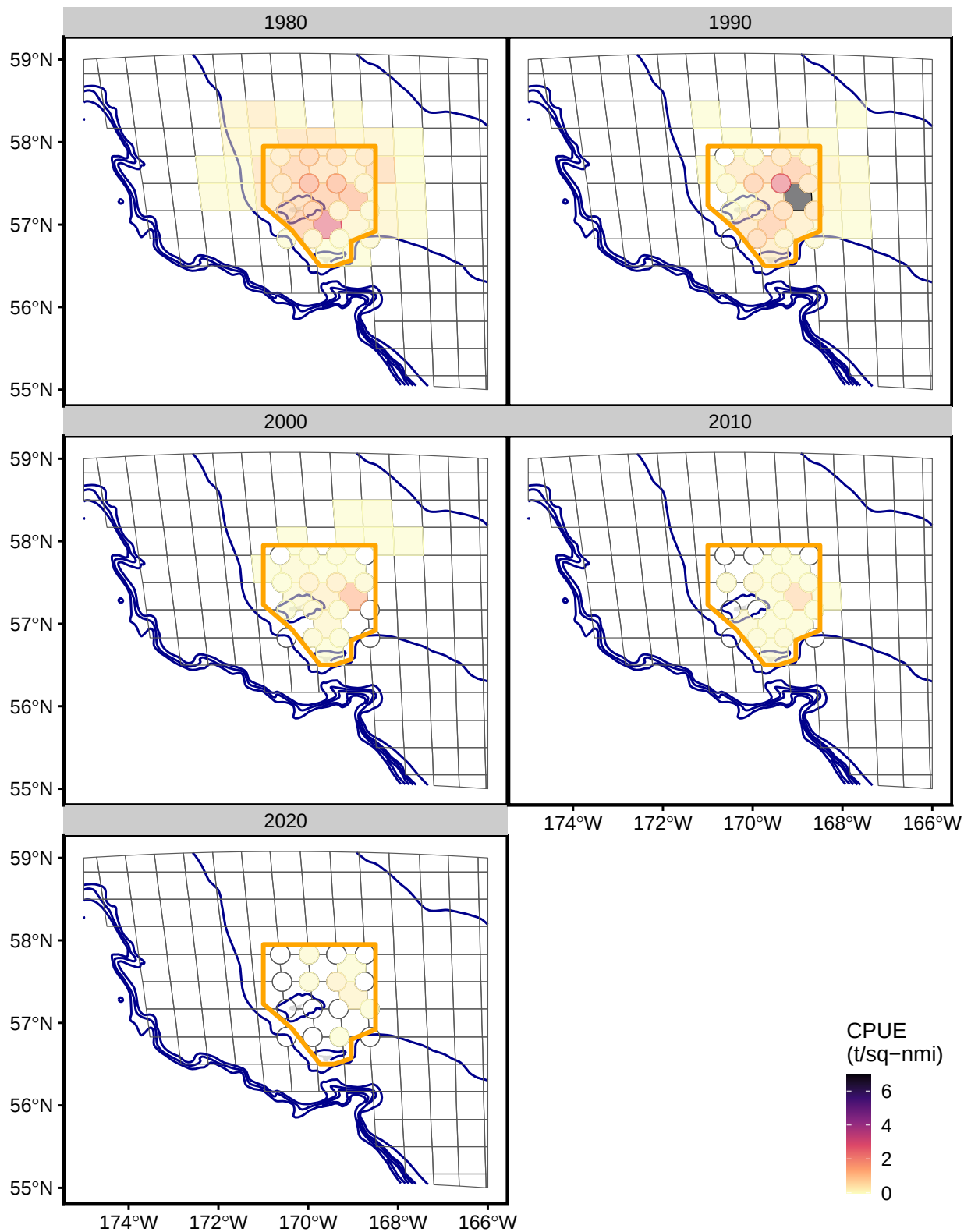


Figure 19. Decadal-average biomass CPUE (t/sq-nmi) by for male PIBKC in the NMFS EBS trawl survey.

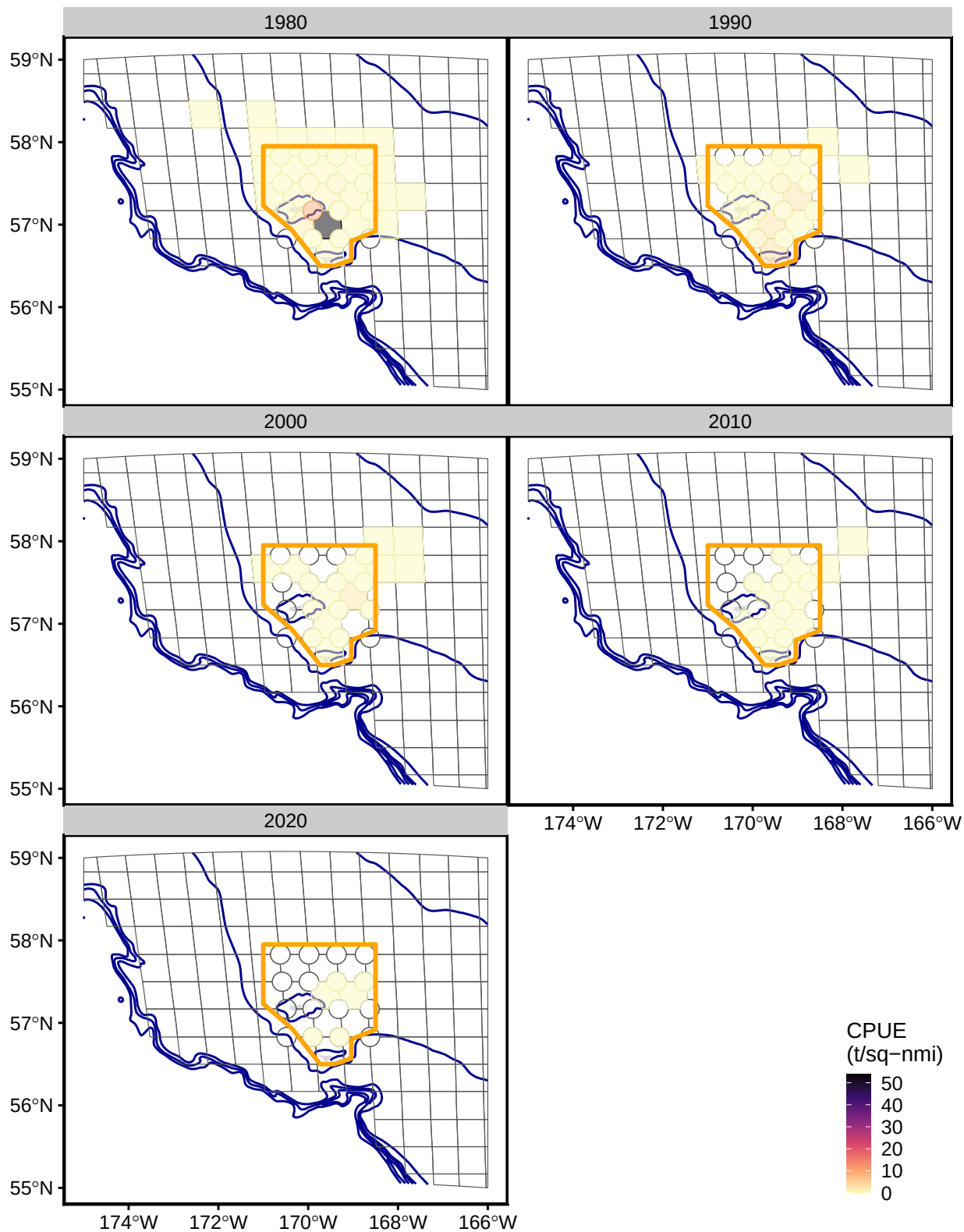


Figure 20. Decadal-average biomass CPUE (t/sq-nmi) by for female PIBKC in the NMFS EBS trawl survey.

References

- Foy, R.J. 2013. 2013 Stock Assessment and Fishery Evaluation Report for the Pribilof Islands blue king crab fisheries of the Bering Sea and Aleutian Islands regions. *In* Stock Assessment and Fishery Evaluation Report for the KING AND TANNER CRAB FISHERIES of the Bering Sea and Aleutian Islands regions 2013 final crab SAFE. North Pacific Fishery Management Council, Anchorage, AK. Available from <https://www.npfmc.org/wp-content/PDFdocuments/resources/SAFE/CrabSAFE/CrabSAFE2013.pdf>.
- Lauth, R.R., and Nichol, D.G. 2013. Results of the 2012 eastern Bering Sea continental shelf bottom trawl survey of groundfish and invertebrate resources. U.S. National Oceanic; Atmospheric Administration (NOAA). Available from <https://apps-afsc.fisheries.noaa.gov/Publications/AFSC-TM/NOAA-TM-AFSC-256.pdf>.
- Stauffer, G.A. 2004. NOAA protocols for groundfish bottom trawl surveys of the Nation's fishery resources. U.S. National Oceanic; Atmospheric Administration (NOAA).
- Zacher, L.S., Richar, J.I., Fedewa, E.J., Ryznar, E.R., and Litzow, M.A. 2023. The 2022 eastern and northern Bering Sea continental shelf trawl surveys: Results for commercial crab species. U.S. National Oceanic; Atmospheric Administration (NOAA). doi:<https://doi.org/10.25923/r6px-9707>.
- Zacher, L.S., Richar, J.I., and Foy, R.J. 2020. The 2019 eastern and northern Bering Sea continental shelf trawl surveys: Results for commercial crab species. U.S. National Oceanic; Atmospheric Administration (NOAA). doi:<https://doi.org/10.25923/8jdb-5p39>.